CS 305: Computer Networks

Fall 2025

Lecture 4： Application Layer

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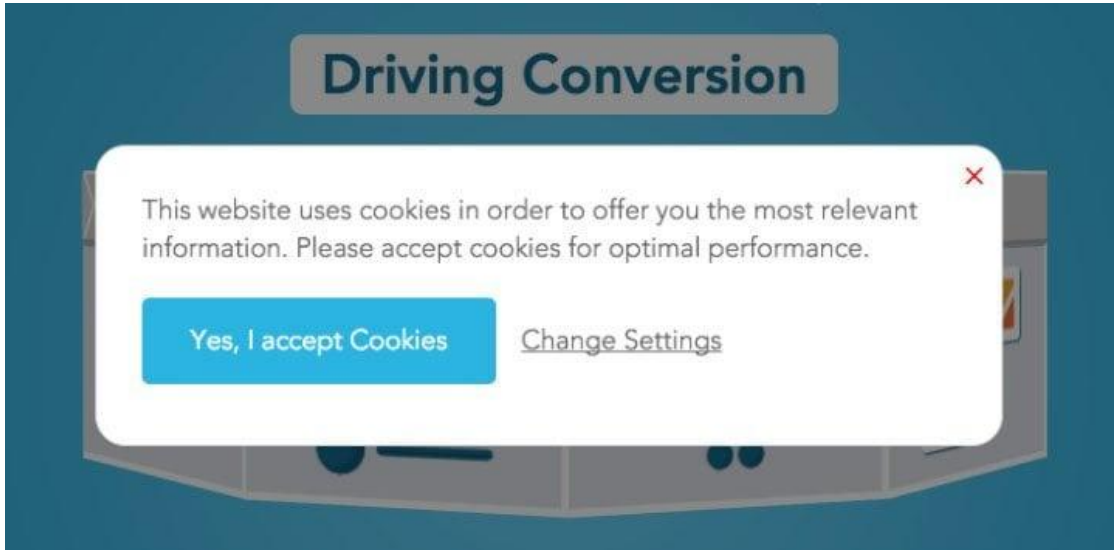
HTTP Outline

- HTTP Overview
  - HTTP runs over TCP
  - HTTP is stateless
  - Persistent and non-persistent connection
- Request and response messages
- Cookies
- Web caching

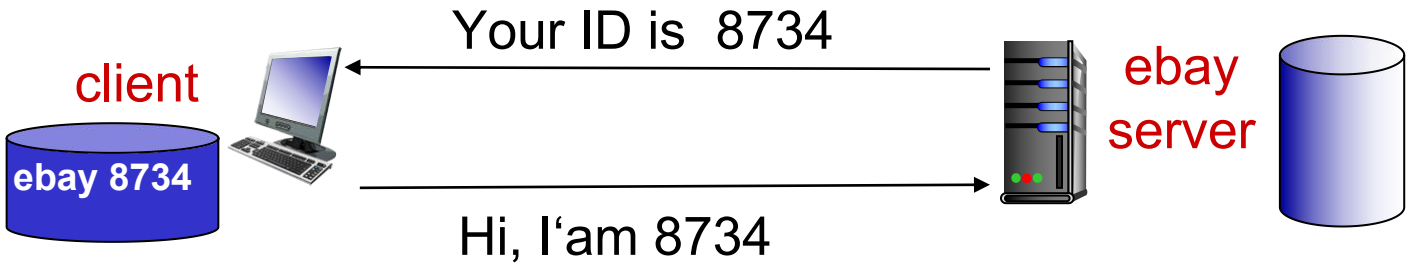


User-server state: cookies

HTTP is Stateless, and servers handle thousands of simultaneous TCP connections.

However, it is often desirable for a Web server to identify users.

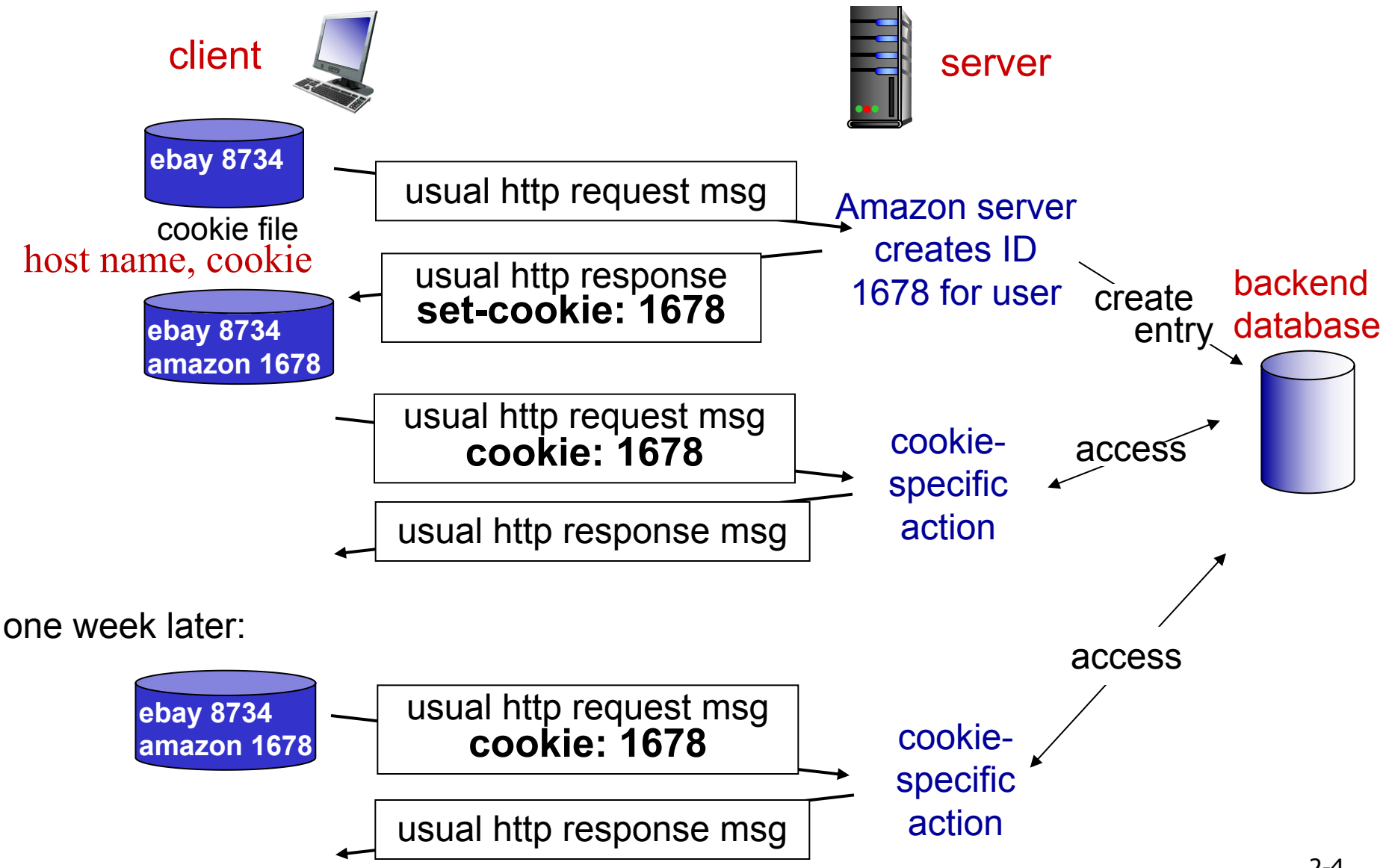
- **Web servers use cookies**



Four components:

1) cookie header line of HTTP *response* message
2) cookie header line in next HTTP *request* message
3) cookie file kept on user's host, managed by user's browser
4) back-end database at Web server

Cookies: keeping "state" (cont.)



Cookies (continued)

- Cookies are associated with web browser
- If Susan also registers herself with Amazon, the database can associate Susan's name with her identification number (cookies).

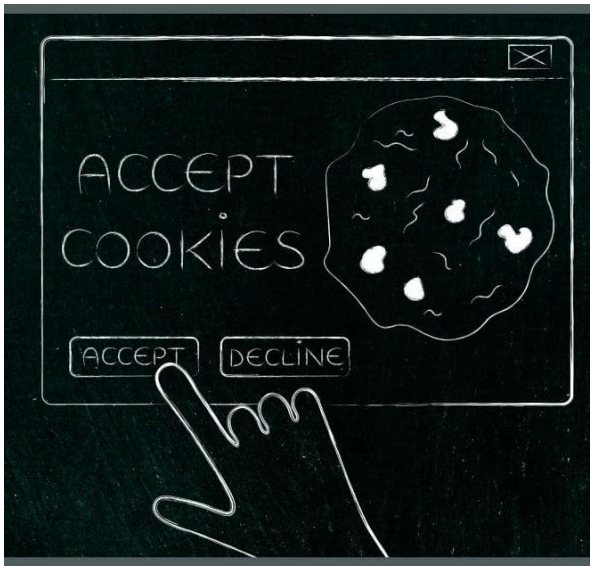
What cookies can be used for:

- authorization
- shopping carts
- recommendations
- user session on top of stateless HTTP

aside

cookies and privacy:

- cookies permit sites to learn a lot about you
- you may supply name and e-mail to sites



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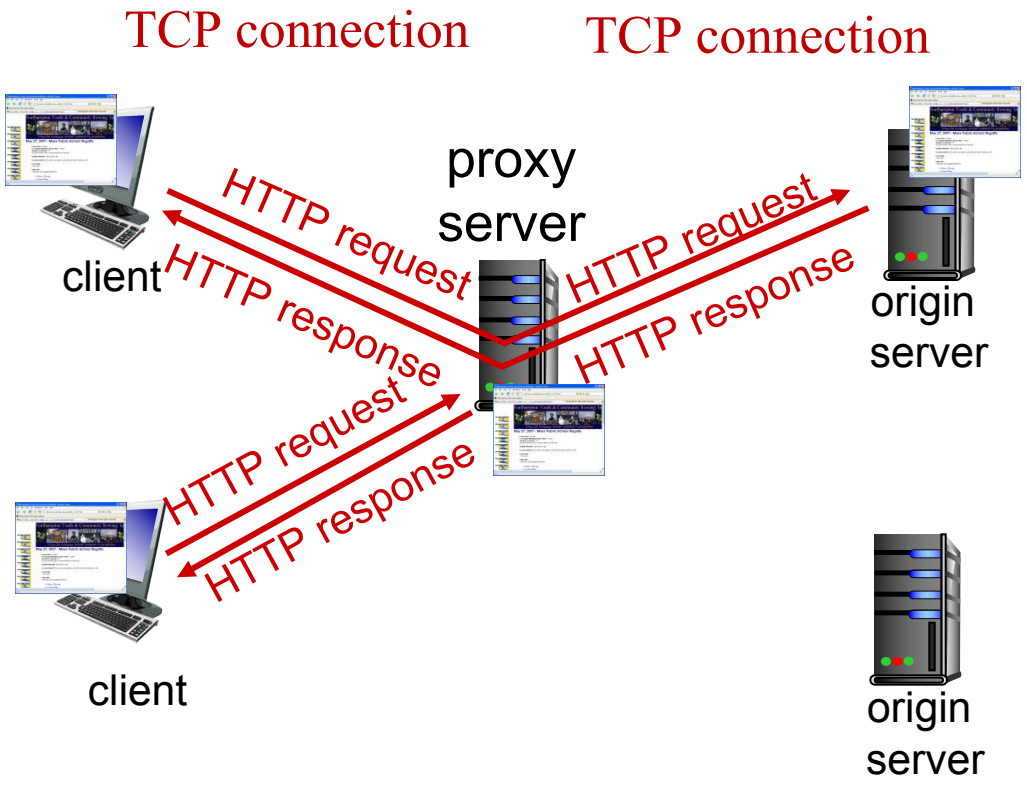
Web caches: proxy （代理） server

goal: satisfy client request without involving origin server

Browser sends all HTTP requests to cache

- object in cache: cache returns object
- else cache requests object from origin server, then returns object to client



More about Web caching

- Cache (Proxy server) acts as both client and server
  - server for original requesting client
  - client to origin server
- typically cache is installed by ISP (university, company, residential ISP)

Why Web caching?

- reduce response time for client request (bottleneck bandwidth)
- reduce traffic on an institution's access link
- Internet dense with caches: enables "poor" content providers to effectively deliver content (so too does P2P file sharing)

Caching example:

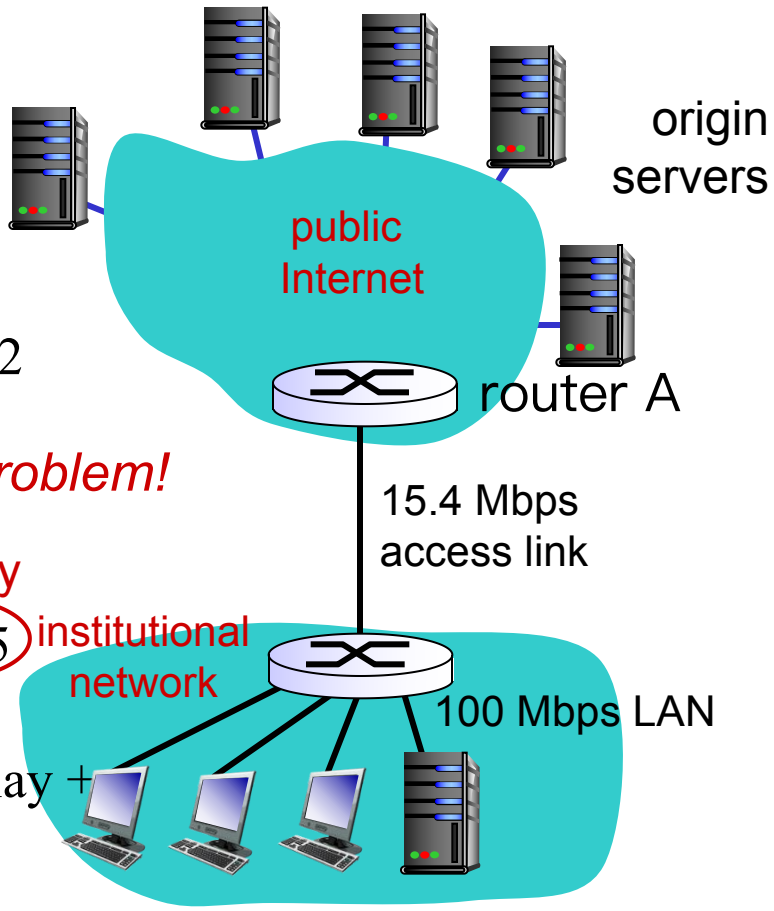
Assumptions:

- avg object size: 1M bits
- avg request rate from browsers to origin servers:15 requests/sec
- avg data rate to all browsers: 15 Mbps
- RTT from router A to any origin server: 2 sec → "Internet delay"
- access link rate: 15.4 Mbps

Consequences:

- LAN utilization: 15Mbps/100Mbps=0.15
- access link utilization =15/15.4= 0.974
- total delay = Internet delay + access delay + LAN delay

  = 2 sec + minutes + milliseconds



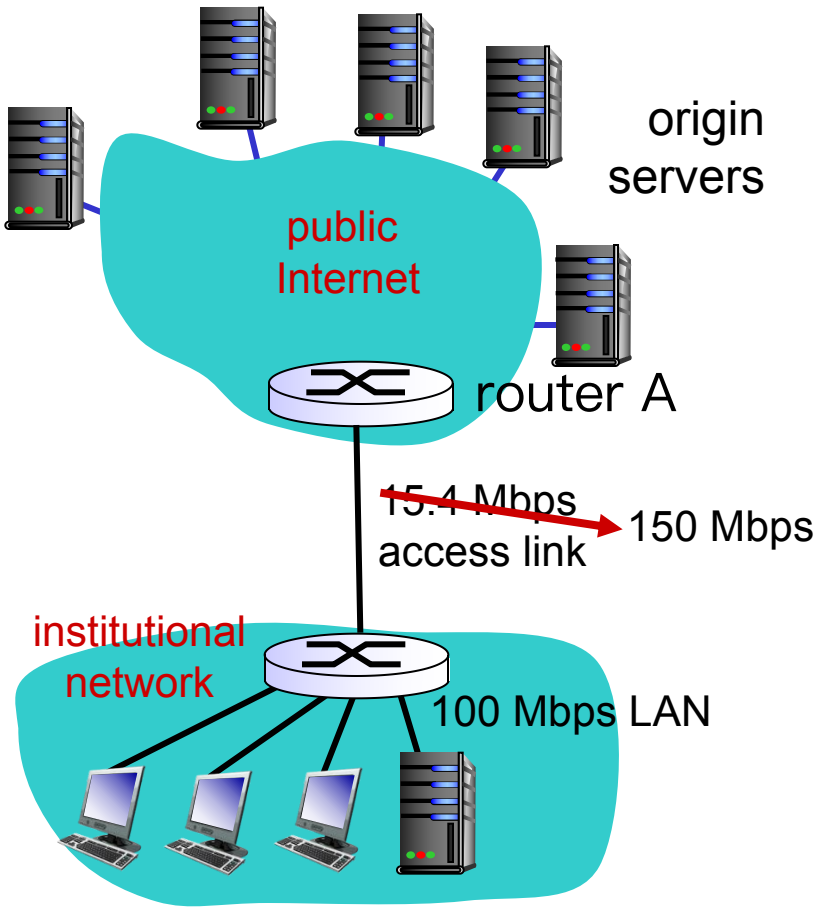
Caching example: fatter access link

assumptions:

- avg object size: 1M bits
- avg request rate from browsers to origin servers:15/sec
- avg data rate to browsers: 15 Mbps
- RTT from router A to any origin server: 2 sec
- access link rate: 15.4 Mbps

consequences:

- LAN utilization: 0.15
- access link utilization = ~~0.974~~ 0.1
- total delay = Internet delay + access delay + LAN delay
  = 2 sec + ~~minutes~~ + milliseconds
  milliseconds



Cost: increased access link speed (not cheap!)

Caching example: install local cache

assumptions:

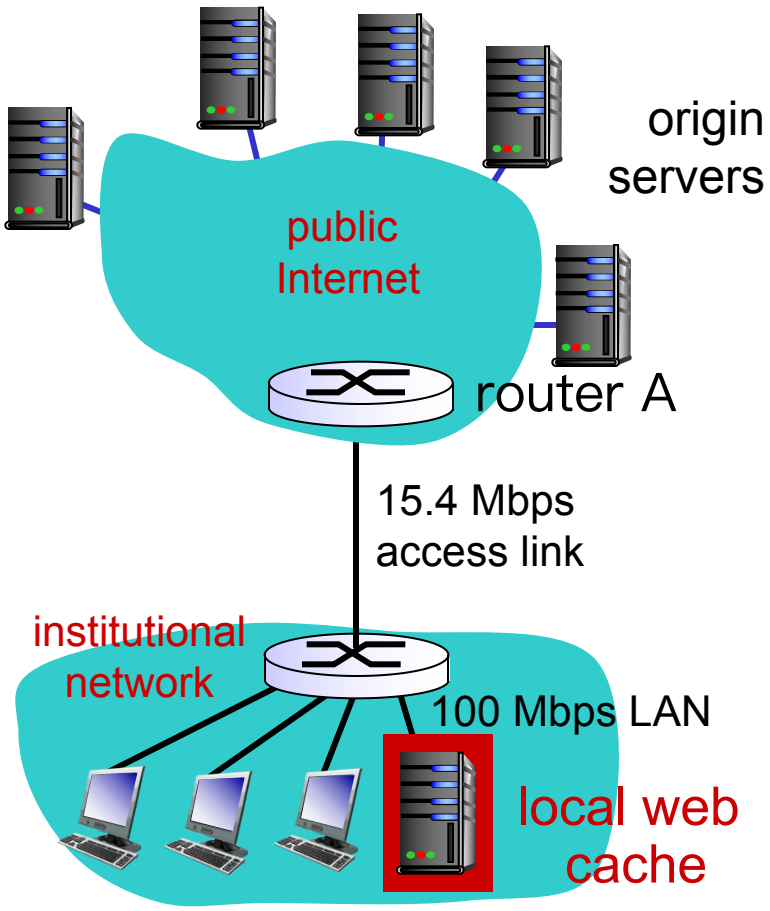
- avg object size: 1M bits
- avg request rate from browsers to origin servers:15/sec
- avg data rate to browsers: 15 Mbps
- RTT from router A to any origin server: 2 sec
- access link rate: 15.4 Mbps

consequences:

- LAN utilization: 0.15
- access link utilization = ?
- total delay = ?

How to compute link utilization, delay?

Cost: web cache (cheap!)



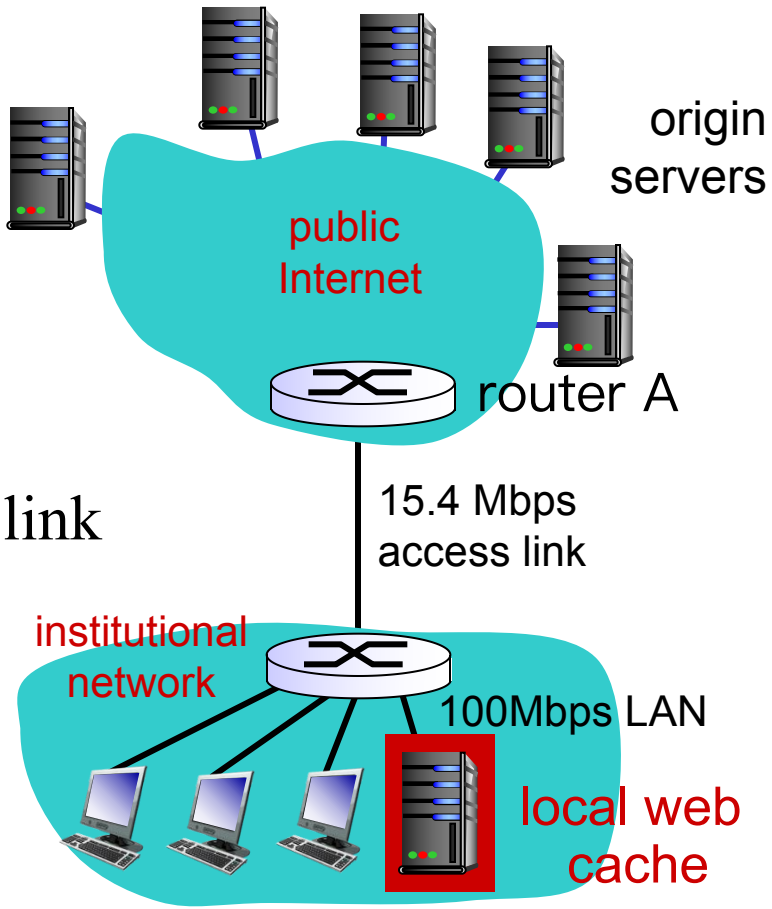
Hit rates: the fraction of requests that are satisfied by a cache. Typically, 0.2—0.7.

Caching example: install local cache

Calculating access link utilization, delay with cache:

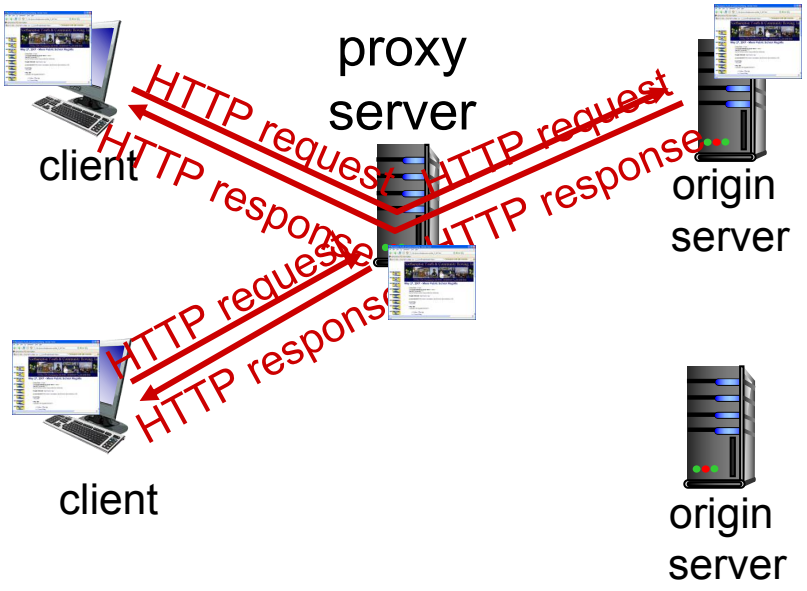
- suppose cache hit rate is 0.4
  - 40% requests satisfied at cache, 60% requests satisfied at origin
- access link utilization:
  - 60% of requests use access link
- data rate to browsers over access link

  = 0.6*15 Mbps = 9 Mbps
  - utilization = 9/15.4 = 0.58
- Average delay
  - = 0.6 * (delay from origin servers) +0.4 * (delay when satisfied at cache)
  - = 0.6 (2.01) + 0.4 (~msecs) = ~ 1.2 secs
  - less than with 150 Mbps link (and cheaper too!)



Typically, a traffic intensity less than 0.8 corresponds to a small delay, say, tens of milliseconds

Conditional GET



The copy of an object residing in the cache may be out-of-date:

Conditional GET

- GET method
- If-Modified-Since

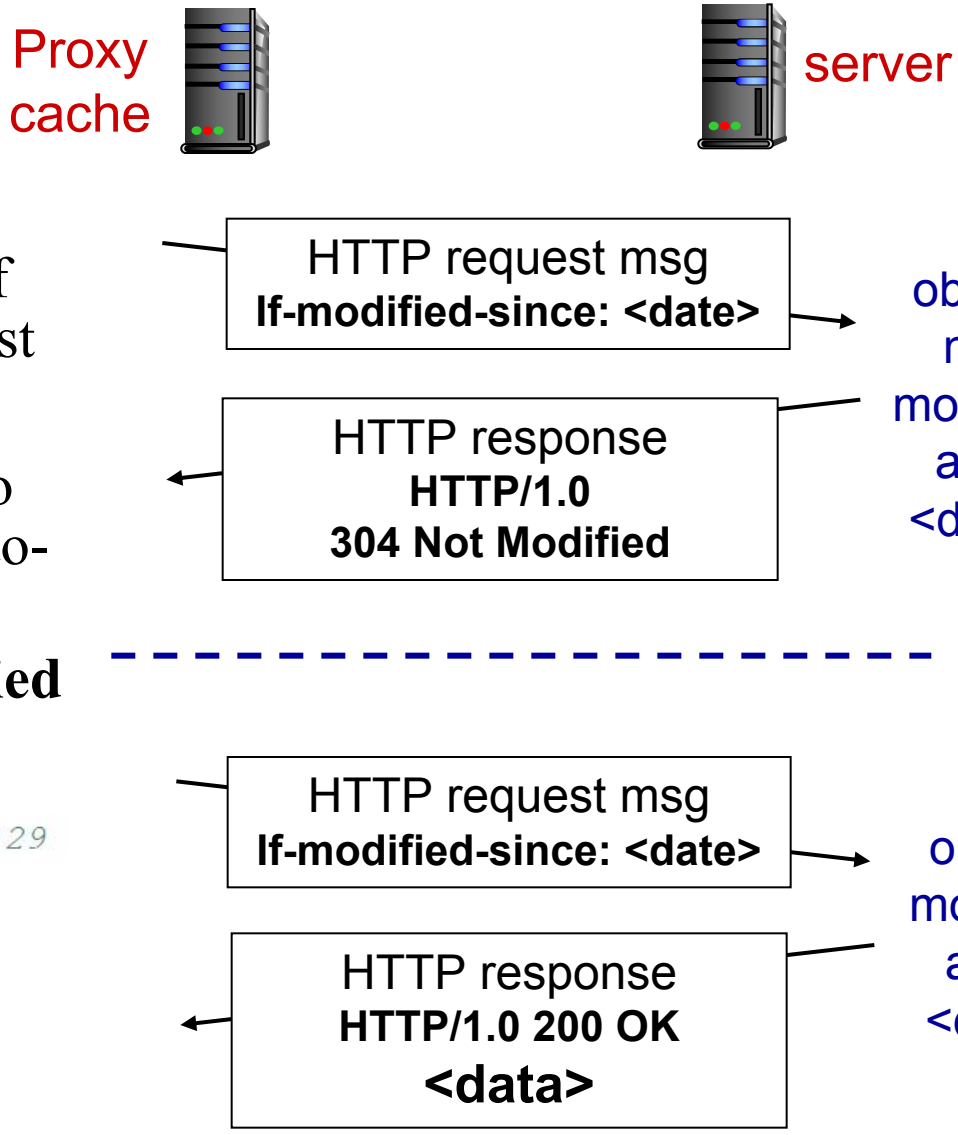
```
GET /fruit/kiwi.gif HTTP/1.1
Host: www.exotiquecuisine.com
If-modified-since: Wed, 9 Sep 2015 09:23:24
```

Goal: allows a cache to verify that its objects are up to date

- don't send object if cache has up-to-date cached version
- no object transmission delay
- lower link utilization

Conditional GET

When a browser requests an object via proxy cache:



- *Proxy cache:* specify date of cached copy in HTTP request

  If-modified-since: <date>

- *Server:* response contains no object if cached copy is up-to-date:

  HTTP/1.0 304 Not Modified

```
HTTP/1.1 304 Not Modified
Date: Sat, 10 Oct 2015 15:39:29
Server: Apache/1.3.0 (Unix)

(empty entity body)
```

HTTP Summary

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Chapter 2: outline

2.1 principles of network applications

2.2 Web and HTTP

2.3 electronic mail

- SMTP, POP3, IMAP

2.4 DNS

2.5 P2P applications

2.6 video streaming and content distribution networks

2.7 socket programming with UDP and TCP

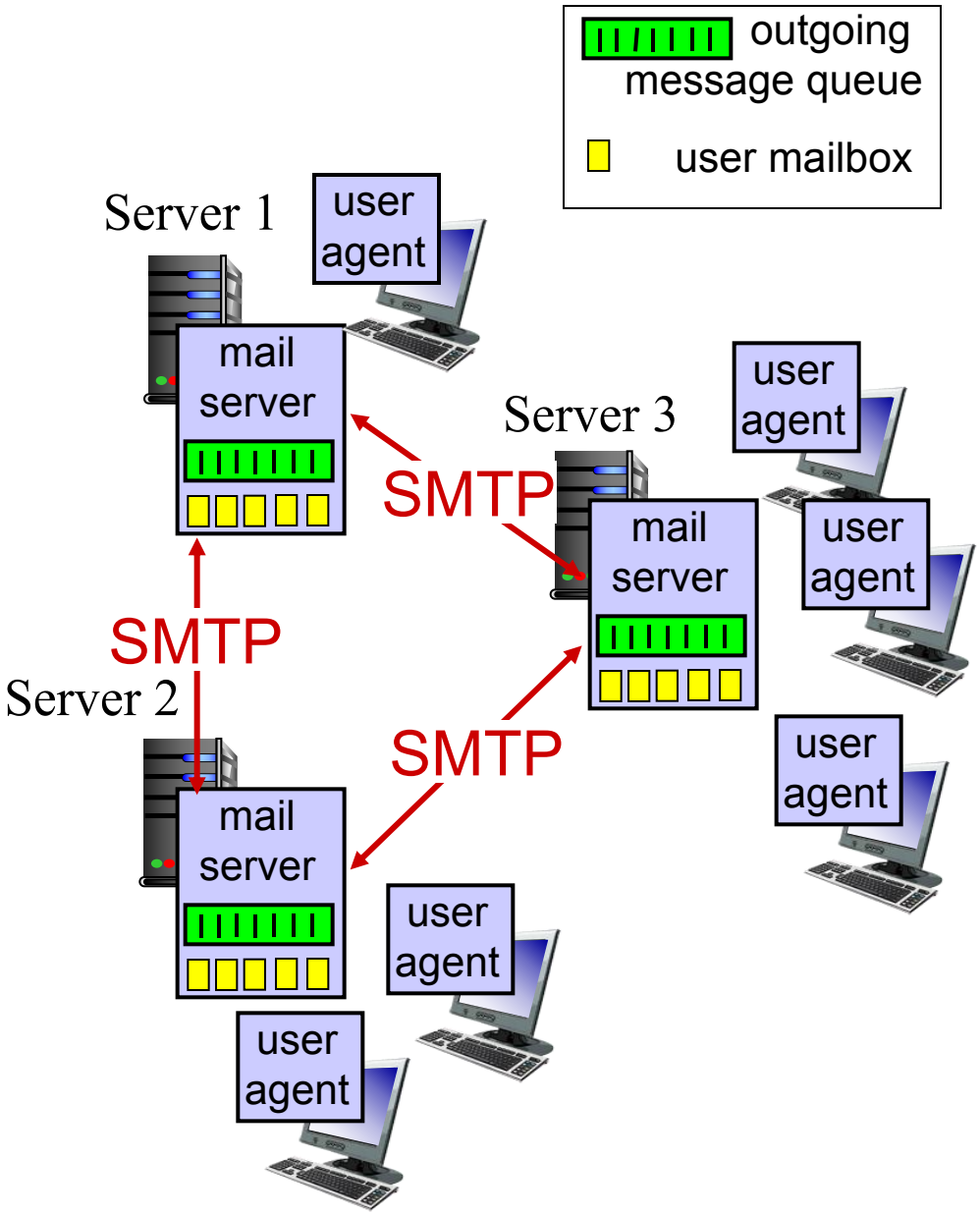
Electronic Mail Overview

- Overview
  - Main components
  - Alice sends an email to Bob
- SMTP
- Mail Message Format
- Mail Access Protocol
  - POP3
  - IMAP
  - HTTP: Web-based Email

Electronic mail

Three major components:

- user agents
- mail servers
- simple mail transfer protocol (SMTP): use TCP

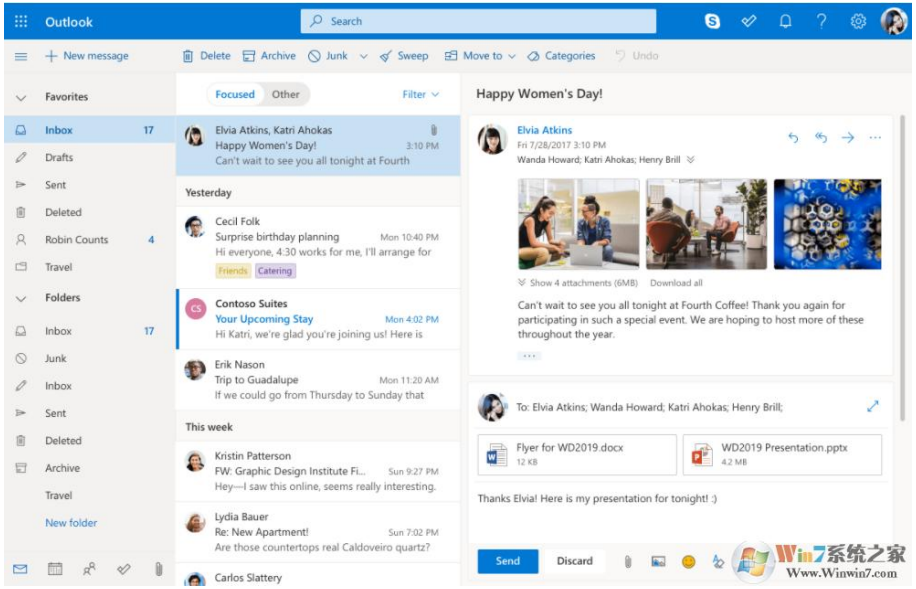


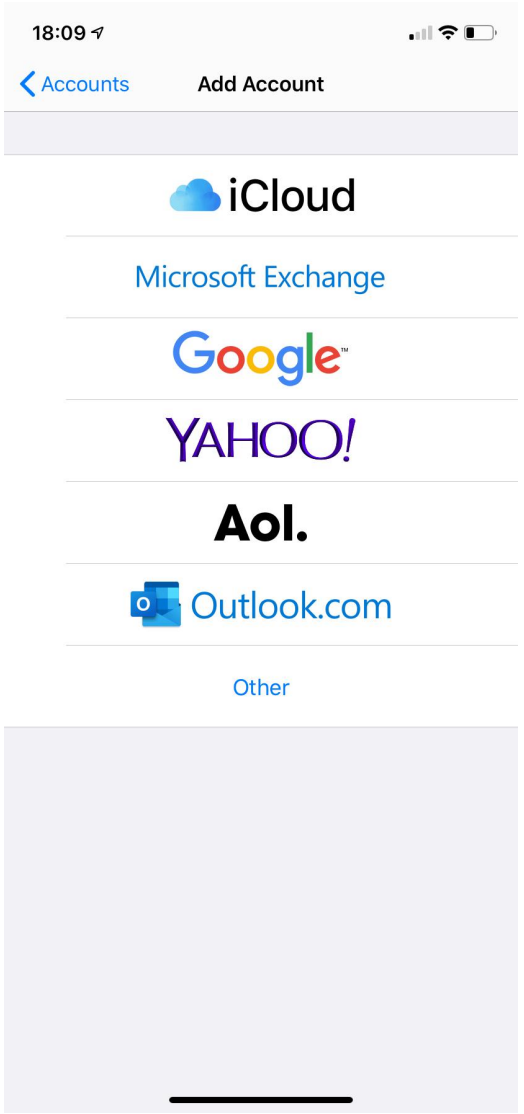
Electronic mail: User Agent

User Agent

邮件客户端，相当于我们和邮件系统的“接口”

- a.k.a. “mail reader”
- Allow users to read, reply to, forward, save and compose messages
- e.g., Outlook, iPhone mail client



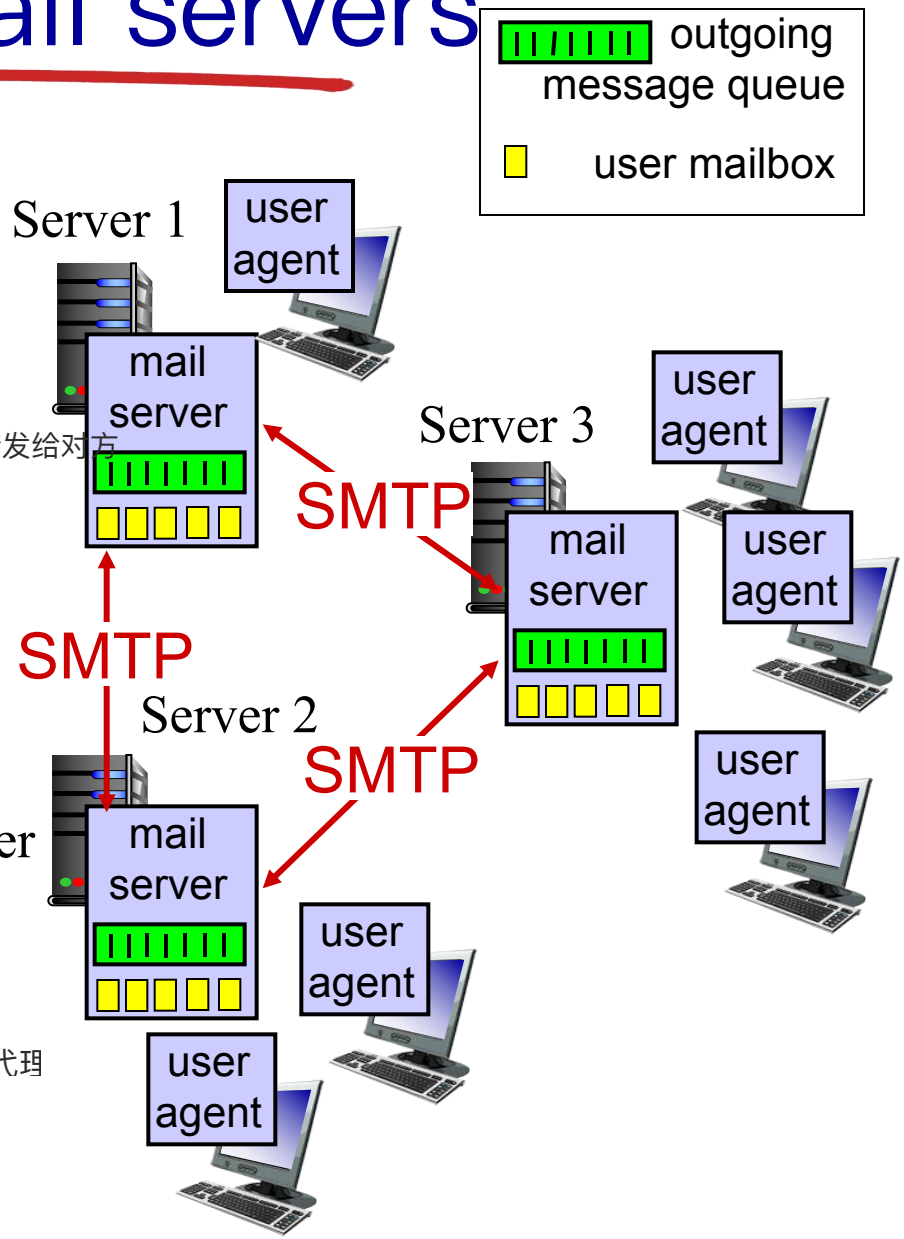


Electronic mail: mail servers

Mail servers:

- Always-on hosts
- *User mailbox* contains outgoing, incoming messages
- *Message queue* of outgoing (to be sent) mail messages
- *Simple Mail Transfer Protocol (SMTP)* between mail servers to send email messages
  - client process: sending mail server
  - server process: receiving mail server

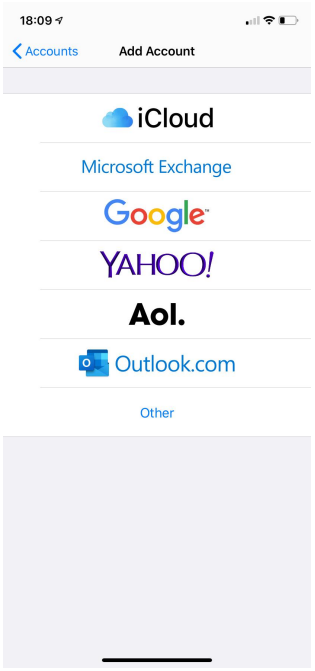
  Both client and server sides of SMTP run on mail server.



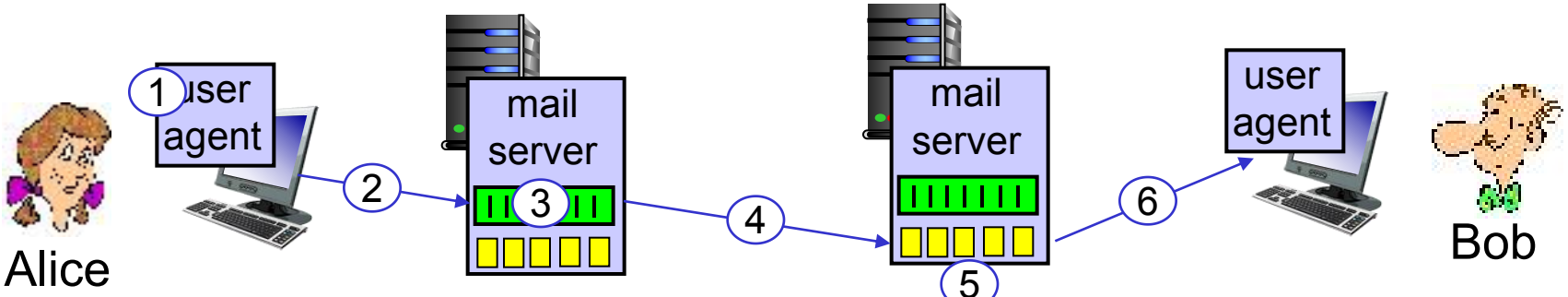
邮件必须先交给服务器才能转发给对方

发邮件时、本地用户代理 → 本地服务器 → 对方服务器 → 对方用户代理

Scenario: Alice sends message to Bob



1) Alice uses user agent to compose message "to" `bob@hamburger.edu`
2) Alice's user agent sends message to her mail server; message placed in message queue
3) client side of SMTP opens TCP connection with Bob's mail server
4) SMTP client sends Alice's message over the TCP connection
5) Bob's mail server places the message in Bob's mailbox
6) Bob invokes his user agent to read message



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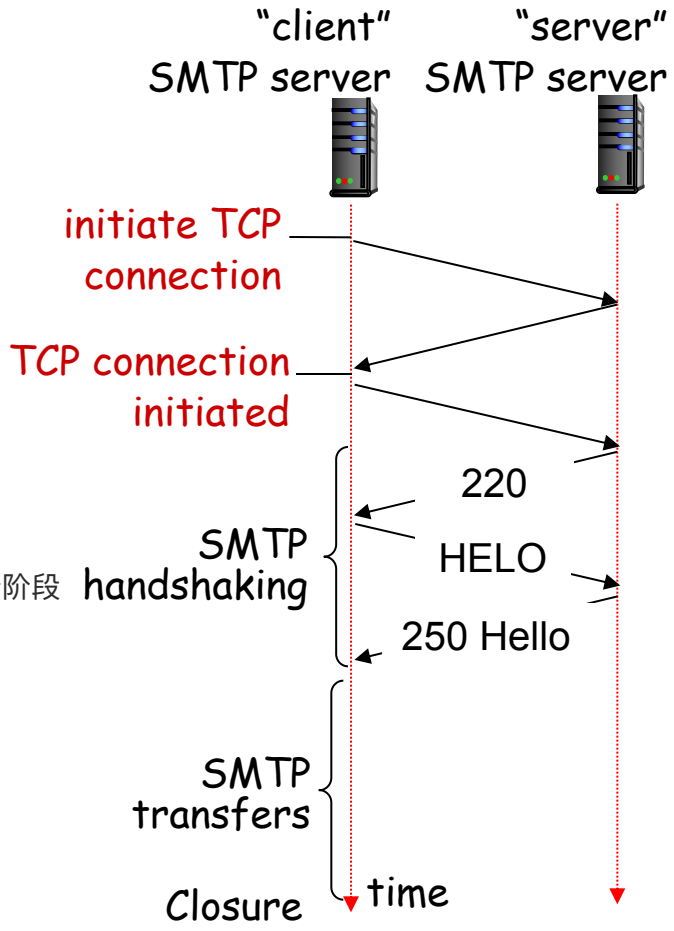
Electronic Mail: SMTP [RFC 2821]

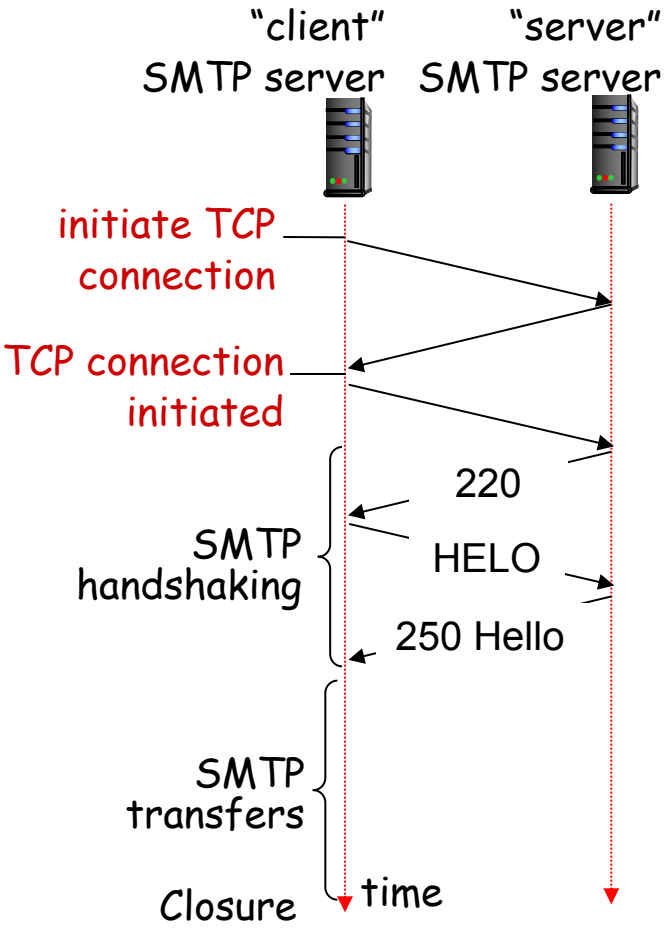
- Uses TCP to reliably transfer email message from client to server, port 25
  - If fail, new attempt after a while (e.g., 30 minutes)
- Direct transfer: sending server to receiving server
  - Direct connection, no intermediate mail server
- Three phases of transfer
  - handshaking (greeting): indicate email address
  - transfer of messages: persistent connection
  - closure

在 TCP 连接建立之后，SMTP 才分为三个阶段



Electronic Mail: SMTP [RFC 2821]

- Two types of messages (like HTTP)
  - commands: text
  - response: status code and phrase
- Entire messages (header & body) must be in ASCII
  - Binary multimedia data → ASCII
  - For HTTP, headers are encoded with ASCII



Sample SMTP interaction

The following are exactly the lines the client (C: crepes.fr) and server (S: hamburger.edu) send after they establishing TCP connections.

commands
response (status code + phrase)

SMTP handshaking

```
S: 220 hamburger.edu
C: HELO crepes.fr
S: 250  Hello crepes.fr, pleased to meet you
```

SMTP transfers

```
C: MAIL FROM: <alice@crepes.fr>
S: 250 alice@crepes.fr... Sender ok
C: RCPT TO: <bob@hamburger.edu>
S: 250 bob@hamburger.edu ... Recipient ok
C: DATA
S: 354 Enter mail, end with "." on a line by itself
C: Do you like ketchup?
C: How about pickles?
C: .
S: 250 Message accepted for delivery
```

Repeat to send multiple messages

Closure

```
C: QUIT
S: 221 hamburger.edu closing connection
```

SMTP: Closing Observations

- SMTP uses persistent connections
- SMTP requires message (header & body) to be in ASCII
- SMTP server uses CRLF.CRLF to determine end of message

回车换行 + 一个点 + 回车换行

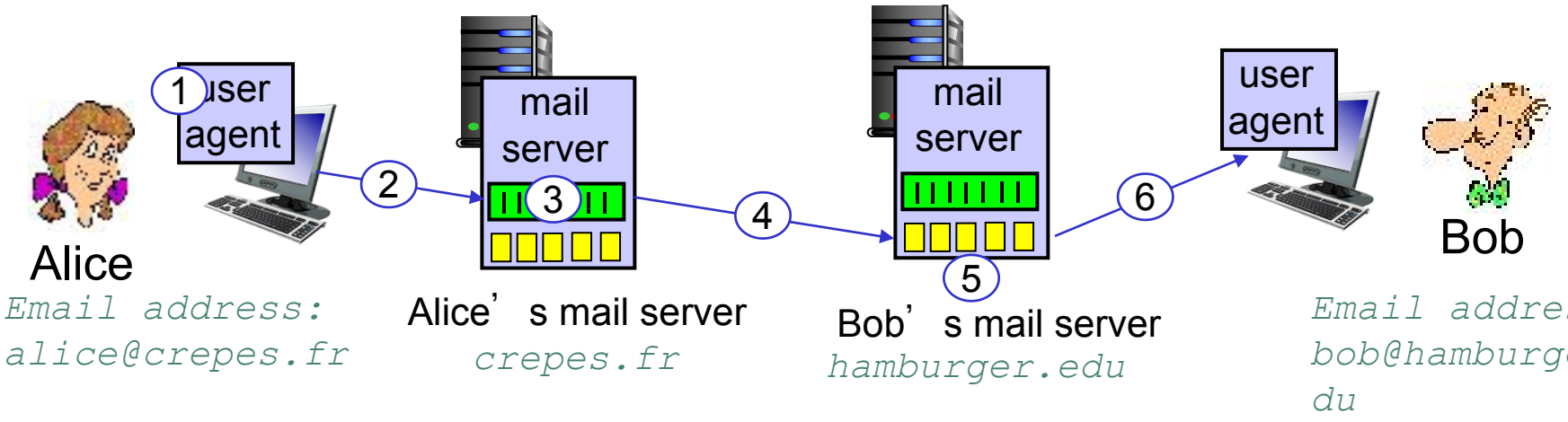
Comparison with HTTP:

- HTTP: pull 客户端主动发请求
- SMTP: push 发送方主动把邮件推给对方

- HTTP: ASCII in header
- SMTP: ASCII in header and body

- HTTP: each object encapsulated in its own response message 每个对象（HTML、图片、JS）都是单独的 respon
- SMTP: multiple objects sent in one message

Alternative Choices?



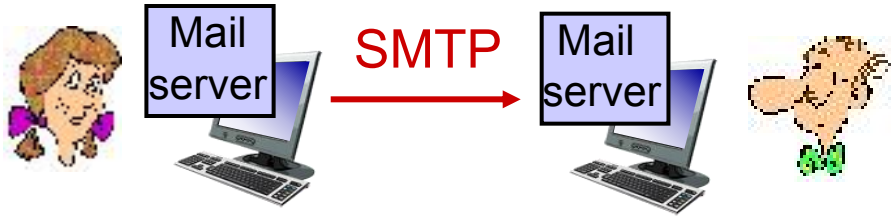
Can we have mail servers directly on user's local PC?

NO

Can we let Alice send to Bob's mail server directly?

NO!

Alternative Choices?



Why not having mail servers directly on user's local PC?

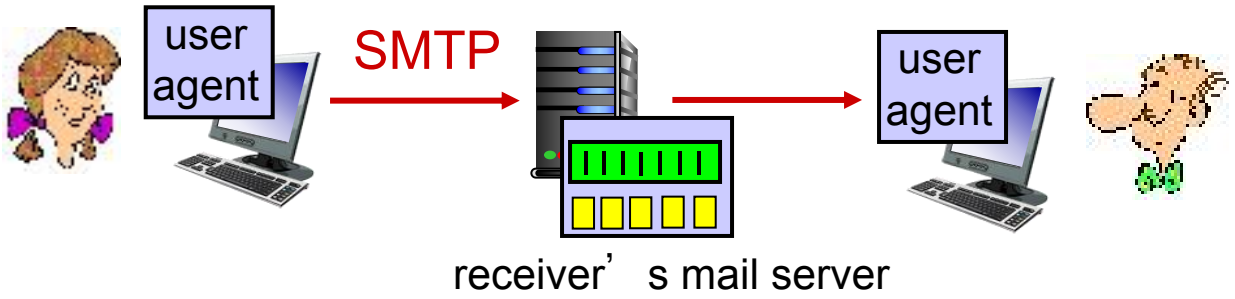
- Recall that a mail server manages mailboxes and runs the client and server sides of SMTP. 负责管理用户邮箱（mailbox），并运行 SMTP 协议的客户端和服务端
- If Bob's mail server were to reside on his local PC, then Bob's PC would have to remain always on in order to receive new mail.

Bob 的 PC 必须 24 小时开机，才能接收新邮件。

一旦电脑关机或掉线，别人就没法把邮件投递给他。

这样对用户不现实，因为个人电脑不可能一直开机、稳定在线。

Alternative Choices?



Why not letting Alice send to Bob’s mail server directly?

- Bob’s mail sever may fail; need to repeatedly send the message until success.

问题：

Bob 的电脑（或他的“服务器”）可能临时故障或不在线。
如果失败，Alice 必须不停重试，直到成功，这样会大大增加客户端的复杂度。

现实做法：

由 接收方的邮件服务器（例如 bob@hamburger.edu
对应的服务器）来接收邮件，保证稳定性。
邮件服务器会负责缓存、重试和最终投递，客户端只需要一次发送就行。

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  - POP3
  - IMAP
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Mail message forn

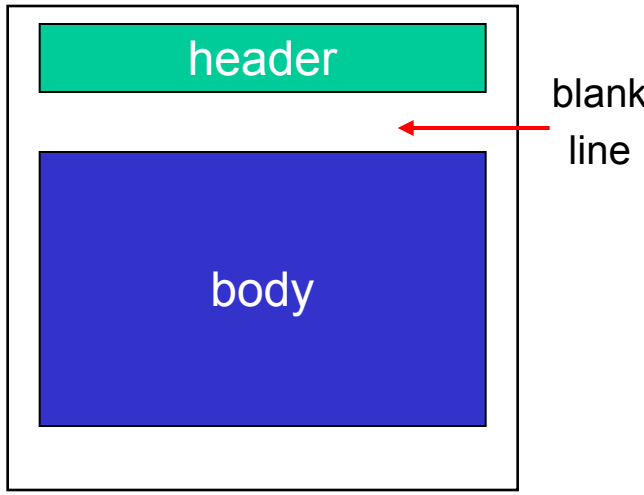
Mail message format (RFC 2822) defines *syntax* for e-mail message itself (like HTML)

- Header lines, e.g.,
  - To:
  - From:
  - Subject:

这些是 邮件内容的一部分，不同于 SMTP 命令里的 MAIL FROM: 和 RCPT TO:

  - these lines are part of the message itself, different from SMTP MAIL FROM:, RCPT TO: commands!

Header 里的 From / To 是邮件文本的一部分，用户会看到



- Body: the “message” , ASCII characters only

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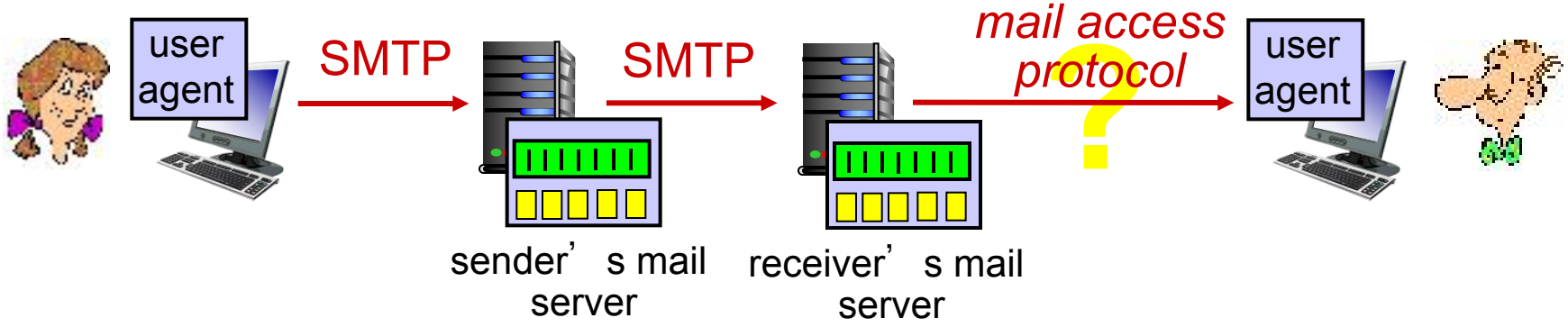
Mail access protocols



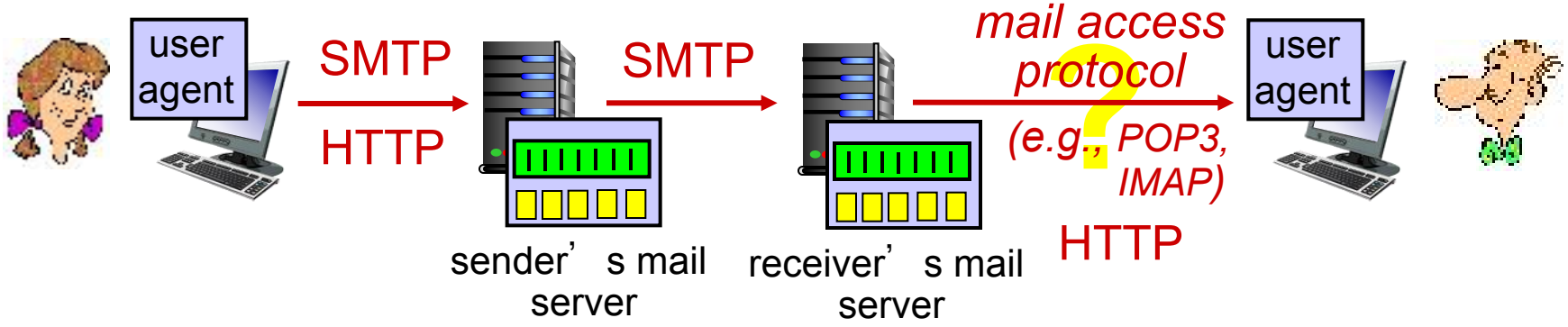
SMTP: delivery to receiver's server

Mail access protocols: How does Bob obtain his message?

SMTP?

No! Because obtaining message is a pull operation.

Mail access protocols



Mail access protocol: retrieval from server

- POP3: Post Office Protocol 3: authorization, download
  - TCP, port 110

用户认证后，从服务器下载邮件到；邮件通常下载后会从服务器删除（除非配置为“保留副本”）。不方便多设备同步（比如手机和电脑同时收信）。

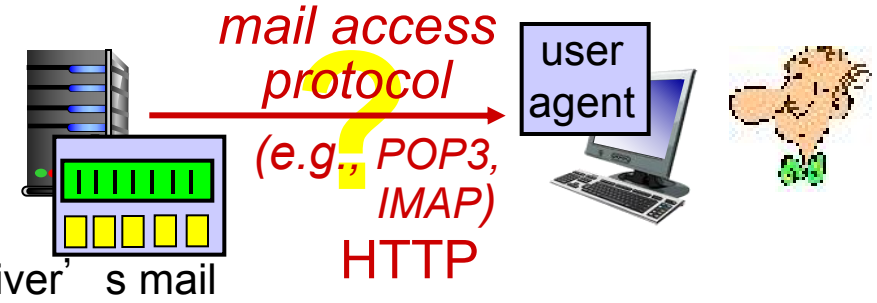
- IMAP: Internet Mail Access Protocol: more features, including maintain folders, keep user state

邮件存储在服务器上，用户可以远程管理（删除、移动、建文件夹）

- HTTP: gmail, Hotmail, Yahoo! Mail, etc.

通过浏览器访问的邮件服务，实际底层还是用 SMTP + IMAP/POP3，但对用户透明。

POP3 protocol



Authorization phase

- client commands:
  - **user:** declare username
  - **pass:** password
- server responses
  - **+OK**
  - **-ERR**

Transaction phase

- client:
  - **list:** list message numbers
  - **retr:** retrieve message by number
  - **dele:** delete
  - **Quit**

Update phase

- After **Quit**, the mail server deletes the messages marked as deletion

```
S: +OK POP3 server ready
C: user bob
S: +OK
C: pass hungry
S: +OK user successfully logged on
```

```
C: list
S: 1 498
S: 2 912
S: .
C: retr 1
S: <message 1 contents>
S: .
C: dele 1
C: retr 2
S: <message 1 contents>
S: .
C: dele 2
C: quit
S: +OK POP3 server signing off
```

Download-and-delete mode

Download-and-keep mode?

keep指是否在服务器上保留

在客户端 quit 之后，服务器才会真正删除之前被 dele 标记的邮件。
所以删除是“延迟生效”的

POP3 (more) and IMAP

More about POP3

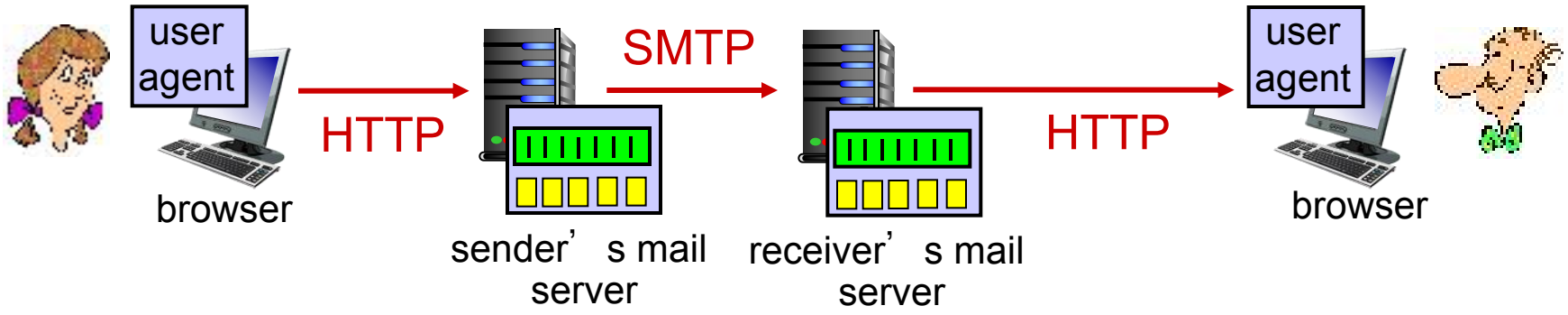
- previous example uses POP3 "download and delete" mode
  - Bob cannot re-read e-mail if he changes client
- POP3 "download-and-keep": reread the message from different machines
- POP3 is stateless across sessions

IMAP

- Maintain a folder hierarchy in one place: at server
- allows user to organize messages in folders
- keeps user state across sessions:
  - names of folders and mappings between message IDs and folder name
- Obtain components of messages 可以只下载邮件头，或附件，按需获取，提高效率

POP3 在不同会话之间不保存用户的状态（比如邮件是否已读/文件夹组织结构）

Web-based Email



Web-based emails are provided by gmail, Hotmail, Yahoo! Mail, etc.

- The user agent is an ordinary web browser

传统模式：
发信：SMTP
收信：POP3 / IMAP

Webmail 模式：
发信/收信：HTTP（浏览器）
服务器间转发：SMTP

Chapter 2: outline

2.1 principles of network applications

2.2 Web and HTTP

2.3 electronic mail

- SMTP, POP3, IMAP

2.4 DNS

2.5 P2P applications

2.6 video streaming and content distribution networks

2.7 socket programming with UDP and TCP

DNS: domain name system

People: many identifiers:

- SSN, name, passport #

Internet hosts, routers:

- hostname, e.g., www.yahoo.com - used by humans
- IP address (32 bit) - used for addressing datagrams

Q: how to map between IP address and name, and vice versa ?

Domain Name System (DNS):

- distributed database implemented in hierarchy of many *name servers*
- application-layer protocol: hosts and name servers communicate to *resolve* names (address/name translation)

DNS Overview

- DNS Services
- DNS Structure
  - Hierarchical structure
  - Iterated and recursive query
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- Inserting records into DNS

DNS Services

- **hostname to IP address translation**
- **host aliasing**
  - 规范主机名 canonical, alias hostnames
  - **www.ibm.com** (alias) is really **servereast.backup2.ibm.com** (canonical)
  - From supplied alias hostname to canonical hostname
- **mail server aliasing**
- **load distribution**
  - replicated Web servers: many IP addresses correspond to one name
  - rotation distributes the traffic (rotate the ordering of IP addresses)

DNS Services

1. An application invokes the client side of DNS
   - specifying the hostname that needs to be translated
2. DNS in the user's host takes over, sending a query message into the network.
   - DNS query and reply messages
   - UDP datagrams to port 53.
3. After a delay, ranging from milliseconds to seconds, DNS in the user's host receives a DNS reply message that provides the desired mapping.
4. The mapping (hostname - IP) is then passed to the invoking application.

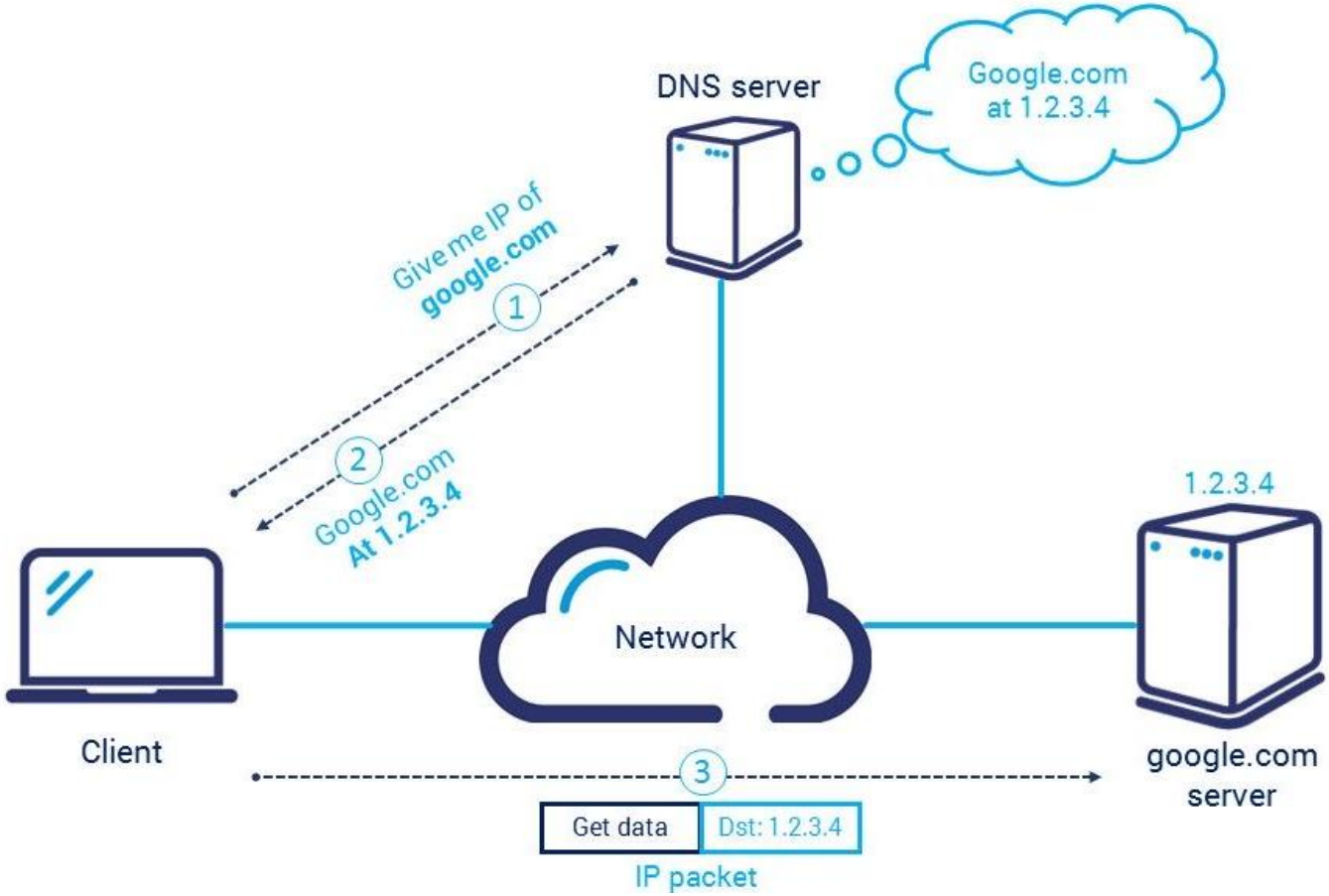
Why UDP?

- fast speed UDP 无连接、无需建立握手，比 TCP 快很多
- smaller data packets

DNS 报文通常很小（512 字节以内），适合用 UDP

DNS Services

From the perspective of the invoking application in the user's host, DNS is a black box providing a simple, straightforward translation service.



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DNS Structure

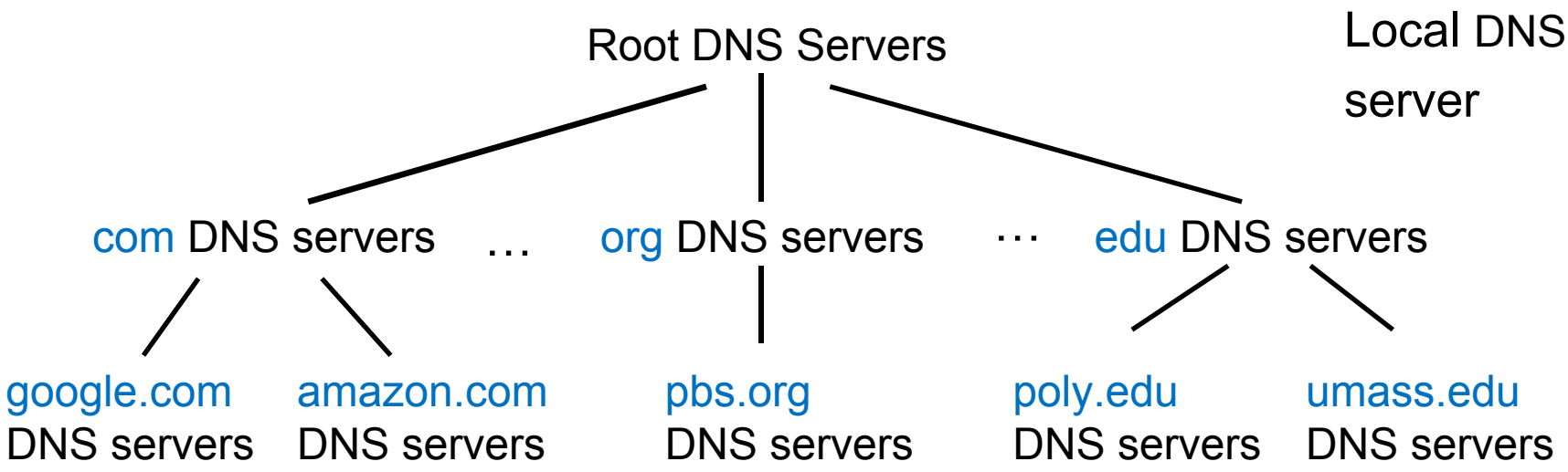
Centralized DNS: 集中式 DNS 意味着所有客户端（全世界的计算机）都把查询请求发给同一个 DNS 服务器。
Clients simply direct all queries to the single DNS server, and the DNS server responds directly to the querying clients.

Why not centralize DNS?

- Single point of failure一旦这个中心服务器宕机，全世界的 DNS 查询都瘫痪
- Traffic volume
- Distant centralized database世界各地的用户访问一个远端服务器，延迟极大
- Maintenance: huge database, update frequently

A: doesn‘t scale!

DNS: a distributed, hierarchical database

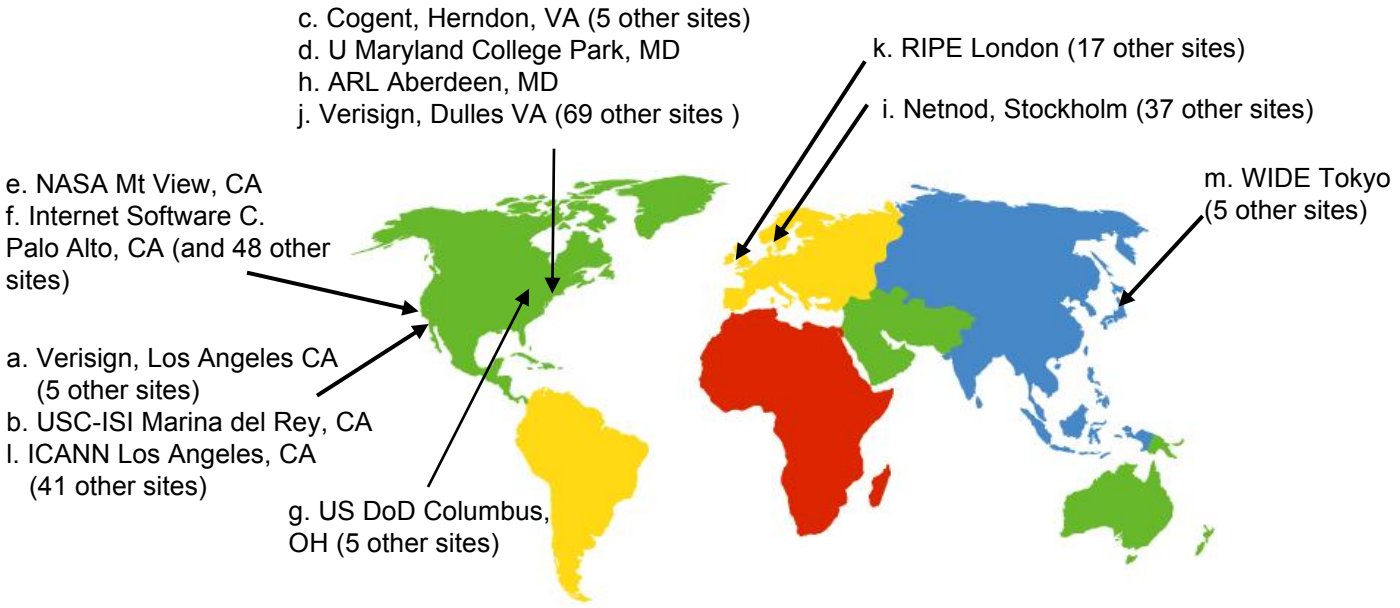


Client wants IP for www.google.com / scholar.google.com:

- Root DNS Servers: find IP address of the .com TLD DNS server
- Top-Level Domain (TLD) DNS: client queries .com DNS server to get google.com authoritative DNS server
- Authoritative DNS servers: client queries google.com DNS server to get IP address for www.google.com / scholar.google.com

DNS: root servers

- Root name server:
  - Provide the IP addresses of the TLD servers



13 logical root name "servers" worldwide

TLD, authoritative servers

Top-level domain (TLD) servers:

- Top-level domains: com, org, net, edu, aero, jobs, museums; top-level country domains: uk, fr, ca, jp
- *Network Solutions* maintains servers for .com TLD
- *Educause* for .edu TLD

Authoritative DNS servers:

- organization's own DNS server(s), providing authoritative hostname to IP mappings for organization's named hosts
- can be maintained by organization or service provider

存放某个组织或域名的“最终答案”——即 hostname ↔ IP 地址的真实映射

Local DNS server

- Does not strictly belong to hierarchy 外部辅助
- Each ISP (residential ISP, company, university) has one
  - also called "default name server"

When a host connects to an ISP, the ISP provides the IP addresses of one or more of local DNS servers

- A host's local DNS server may be typically "close to" the host

When host makes DNS query, query is sent to local DNS server

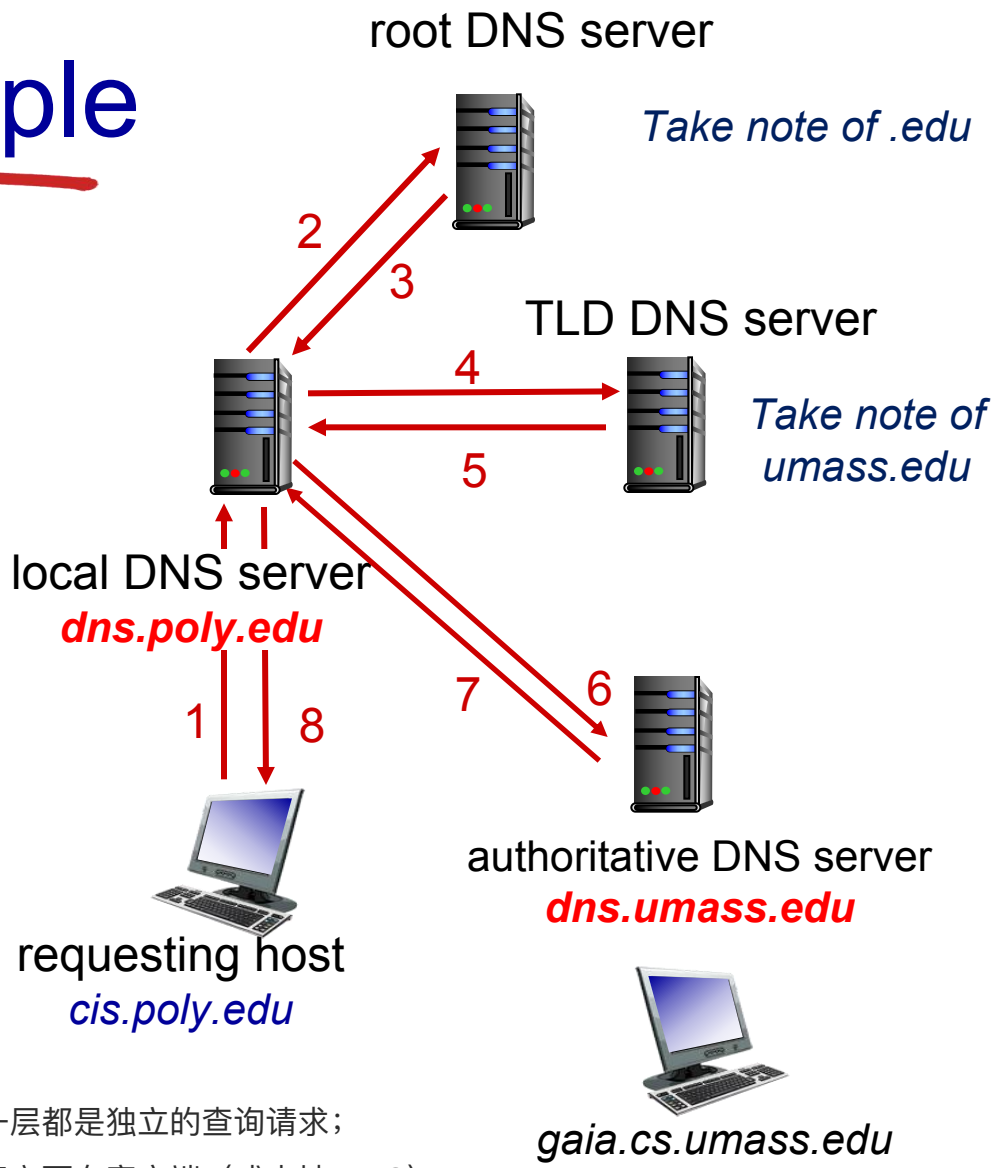
- acts as proxy, forwards query into hierarchy
- has local cache of recent name-to-address translation pairs (but may be out of date!)

1 当用户主机（host）连上网络，ISP 会分配一个或多个本地 DNS 地址。
2 当用户发起域名查询，主机会先问本地 DNS。
3 本地 DNS：
◦ 先查自己的缓存（local cache），如果最近解析过相同域名，就直接返回结果；
◦ 如果缓存里没有，就作为代理（proxy），把查询递交到上层（根 DNS、TLD、权威服务器）；
◦ 最后把结果返回给用户并缓存起来以备下次使用（但缓存可能过期）。

DNS name resolution example

每一级服务器只告诉你“下一个该问谁”，不会帮你查到底

- host at cis.poly.edu wants IP address for **gaia.cs.umass.edu**



Iterated query:

- contacted server replies with the name of another server to contact
- “I don’ t know this name, but ask this server”

查询过程是“多跳式”：每一级都返回“下一步该问谁”

每一层都是独立的查询请求；

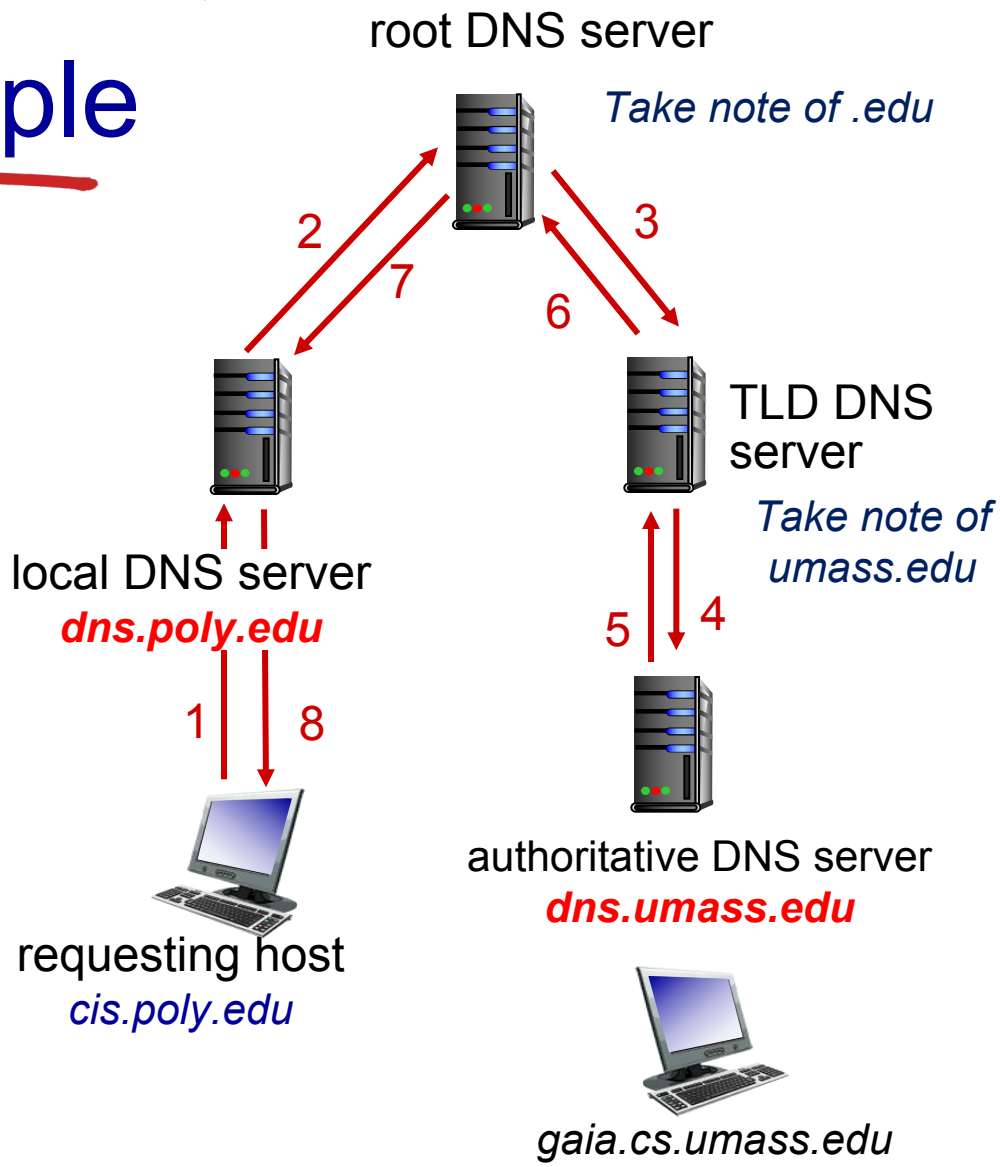
负担主要在客户端（或本地 DNS）；

DNS name resolution example

把整个查询工作“托管”给上层 DNS，让它帮你一路查到底

Recursive query:

- puts burden of name resolution on contacted name server
- heavy load at upper levels of hierarchy?



每一级服务器会替下级“继续向下查”，直到拿到最终答案；

负担主要在上层服务器（尤其是 root / TLD）；

主机只发出一次请求，就能得到结果；

对客户端方便，但上层服务器压力更大。

DNS: caching, updating records

- Once (any) name server learns mapping, it *caches* mapping
  - TLD servers typically cached in local name servers
  - thus root DNS servers not often visited
- Cached entries may be *out-of-date*
  - cache entries timeout (disappear) after some time (e.g., two days)
- Update/notify mechanisms proposed IETF standard
  - RFC 2136

IETF（互联网工程任务组）提出了 DNS 动态更新机制，标准是 RFC 2136

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DNS records

DNS: distributed database storing resource records (RR)

RR format: `(name, value, type, ttl)`

type=A

- **name** is hostname
- **value** is IP address

type=NS

- **name** is domain (e.g., foo.com)
- **value** is hostname of authoritative server for this domain (e.g., dns.foo.com)

type=CNAME

- **name** is alias name for some "canonical" (the real) name
- **www.ibm.com** is really **servereast.backup2.ibm.com**
- **value** is canonical name

type=MX

- **value** is canonical name of the mailserver with **name** (alias name)

DNS records

If a DNS server is authoritative for a particular hostname

- the DNS server will contain a Type A record for the hostname
- (Even if the DNS server is not authoritative, it may contain a Type A record in its cache.)

If a server is not authoritative for a hostname

- the server will contain a Type NS record for the domain that includes the hostname
- it will also contain a Type A record that provides the IP address of the DNS server in the Value field of the NS record.

Example: an .edu TLD server is not authoritative for gaia.cs.umass.edu

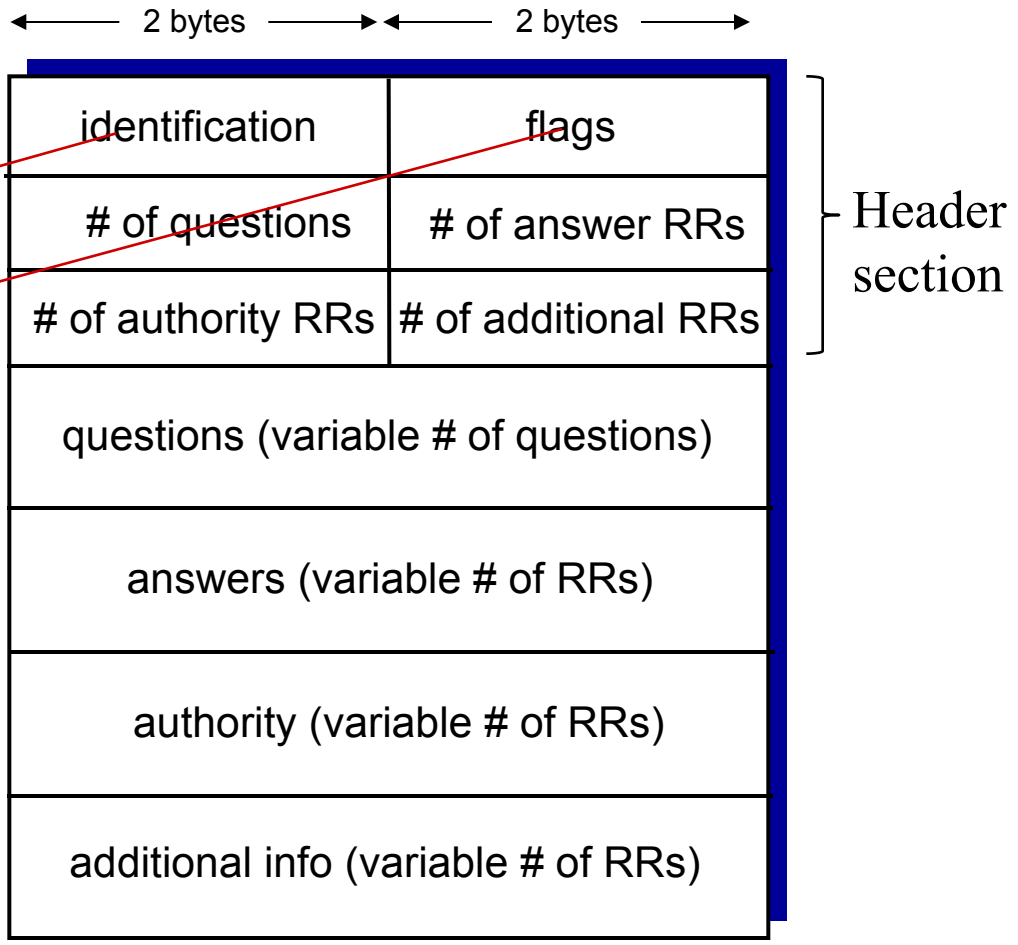
- (umass.edu, dns.umass.edu, NS) .
- (dns.umass.edu, 128.119.40.111, A)

DNS protocol, messages

Query and reply messages, both with same message format

message header

- identification: 16 bit number for query, reply to query uses same number
- flags:
  - query or reply
  - recursion desired
  - recursion available
  - reply is authoritative



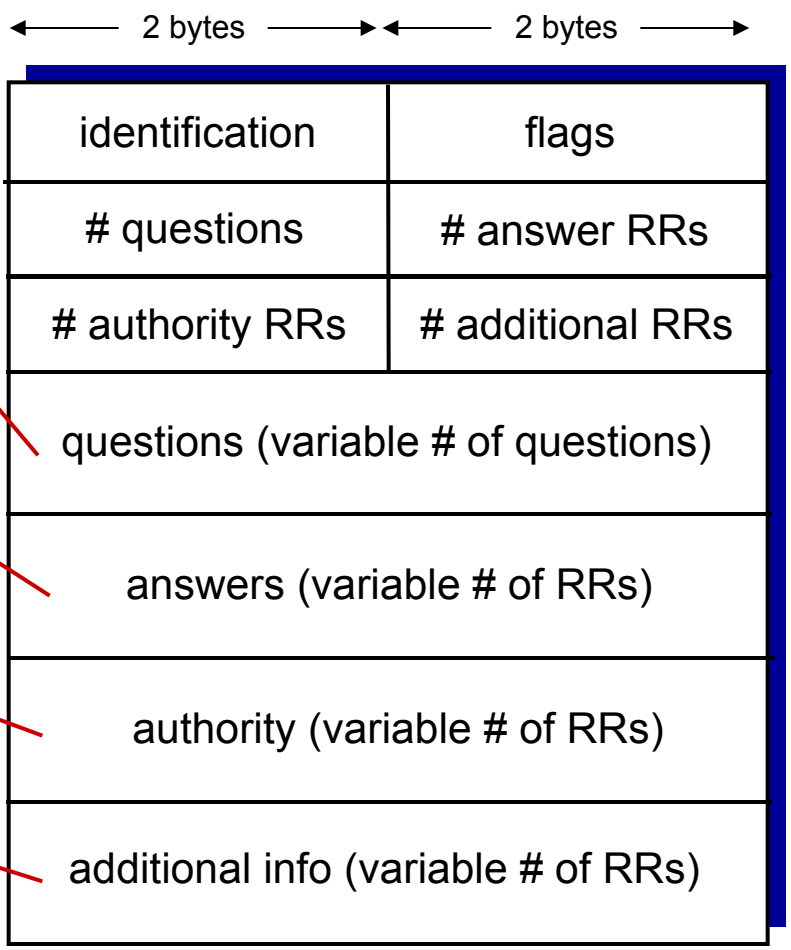
DNS protocol, messages

Name & type fields (e.g., Type A or Type MX)

RRs in response to query (a reply can return multiple RRs)

records of other authoritative servers

additional "helpful" info that may be used



DNS protocol, messages

For example, a reply to **an MX query**

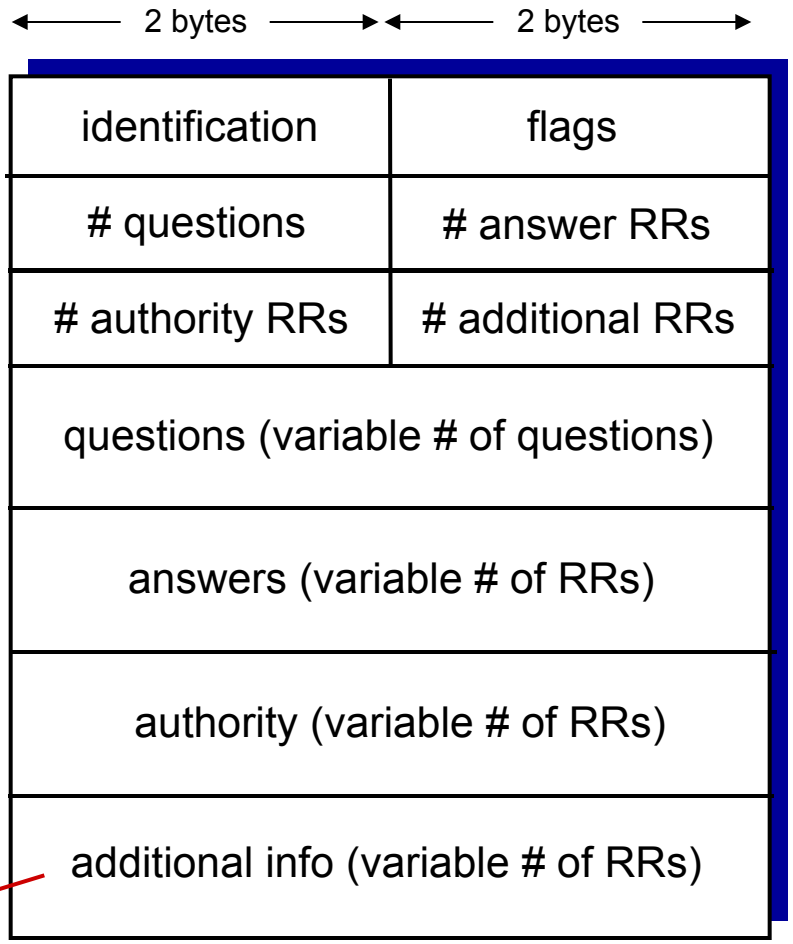
Answer section: Type MX

- an RR providing the canonical hostname of a mail server.

Additional section: Type A

- the IP address for the canonical hostname of the mail server.

additional "helpful" info that may be used



DNS Overview

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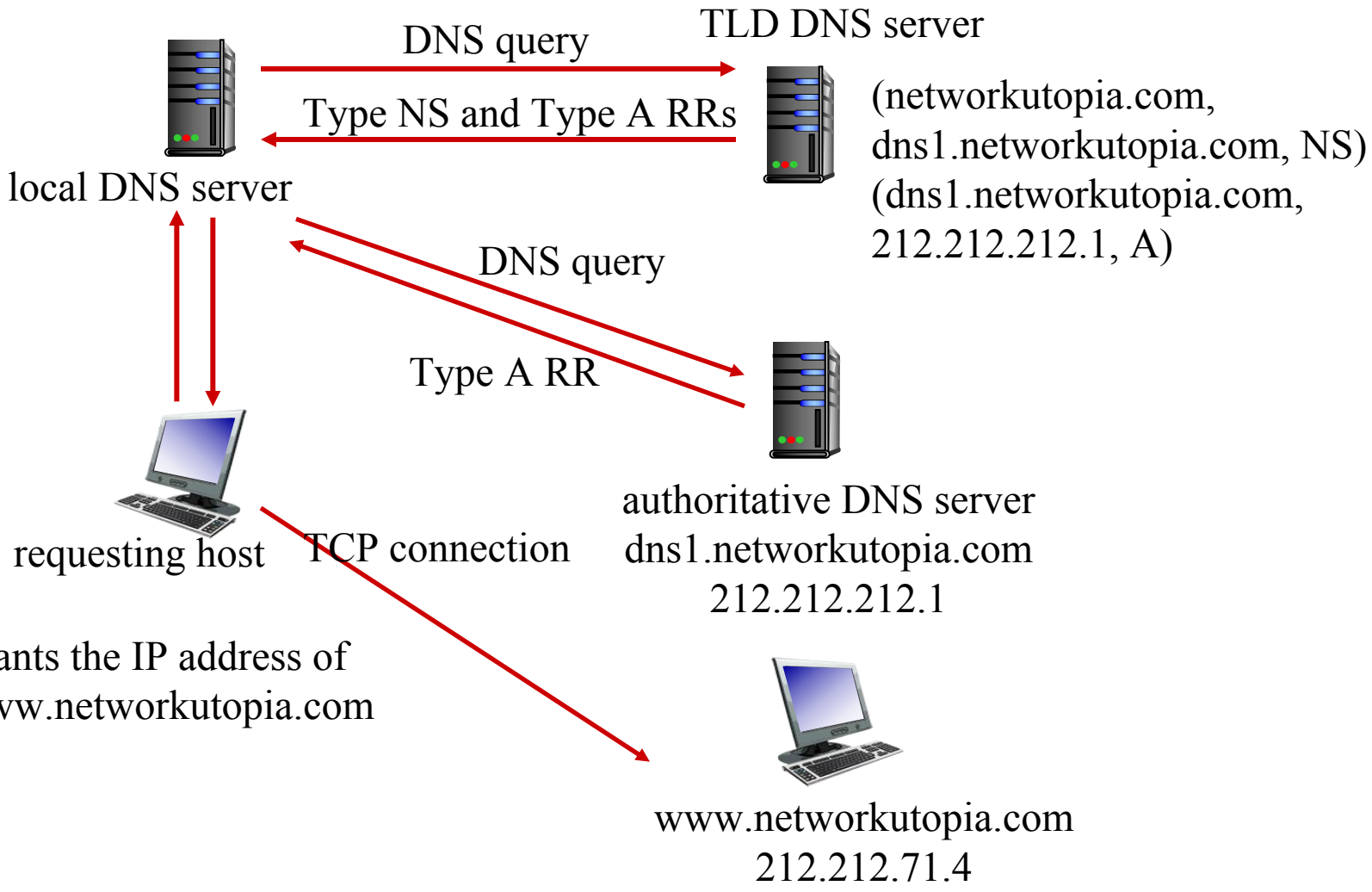
Inserting records into DNS

- Example: new startup "Network Utopia"
- Register name networkuptopia.com at *DNS registrar* (e.g., Network Solutions)
  - provide names, IP addresses of authoritative DNS server (primary and secondary)
  - registrar inserts two RRs into .com TLD server:
    (networkutopia.com, dns1.networkutopia.com, NS)
    (dns1.networkutopia.com, 212.212.212.1, A)

Inserting records into DNS



Attacking DNS

Distributed denial-of-service （DDoS） attacks

- bombard root servers with traffic
  - not successful to date
  - traffic filtering
  - local DNS servers cache IPs of TLD servers, allowing root server bypass
- bombard TLD servers
  - potentially more dangerous

Redirect attacks

- man-in-middle
  - Intercept queries； bogus reply
- DNS poisoning
  - Send bogus replies to DNS server

Exploit DNS for DDoS

- target IP
- Redirect an unsuspecting Web user to attack Web site

Chapter 2: outline

2.1 principles of network applications

2.2 Web and HTTP

2.3 electronic mail

- SMTP, POP3, IMAP

2.4 DNS

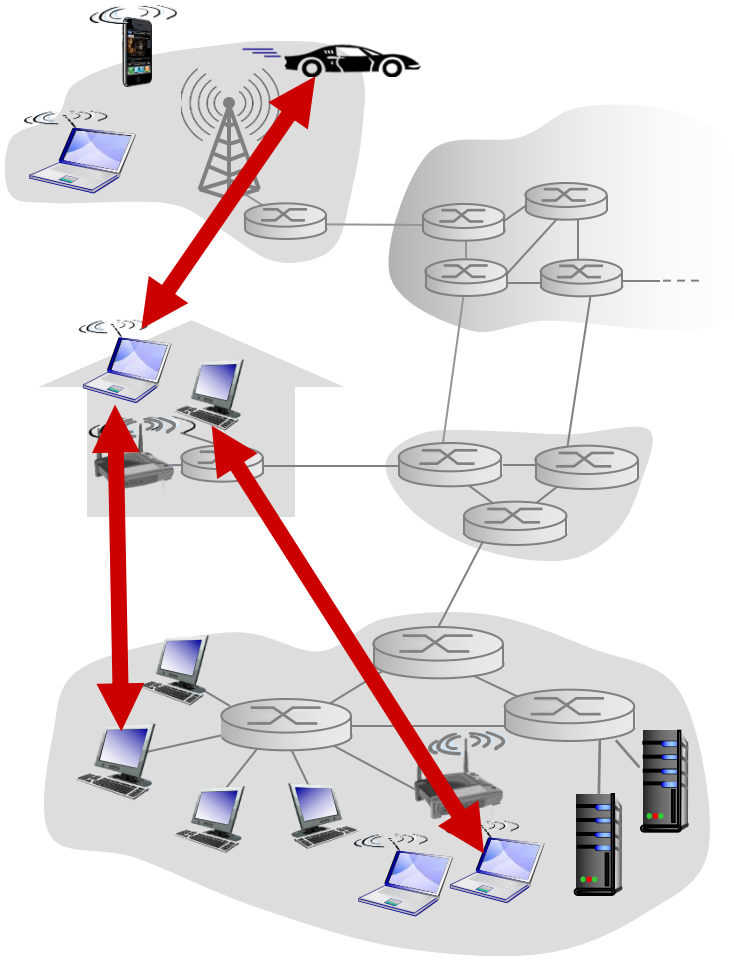
2.5 P2P applications

2.6 video streaming and content distribution networks

2.7 socket programming with UDP and TCP

Pure P2P architecture

- *no* always-on server
- arbitrary end systems directly communicate
- peers are intermittently connected and change IP addresses



Examples:

- file distribution (BitTorrent)
- Streaming (KanKan)
- VoIP (Skype)
- Zoom

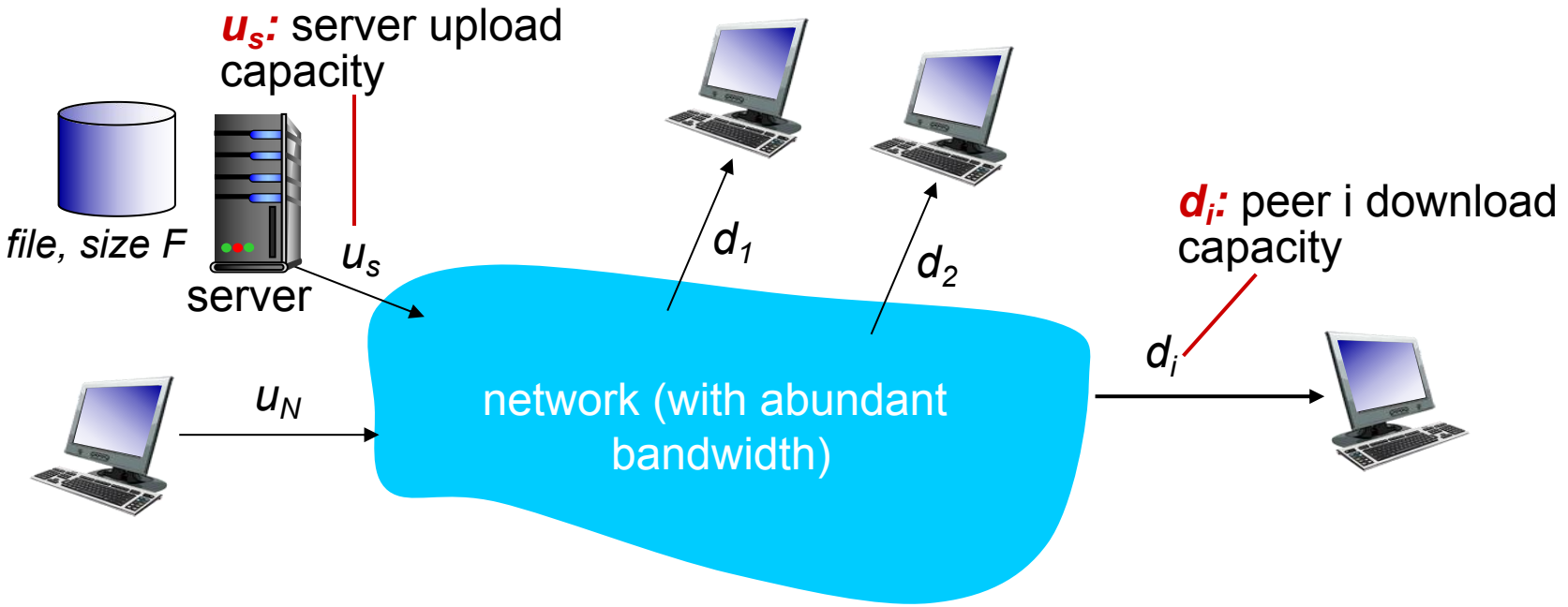
DNS Overview

- P2P vs Client Server
- BitTorrent

File distribution: client–server vs P2P

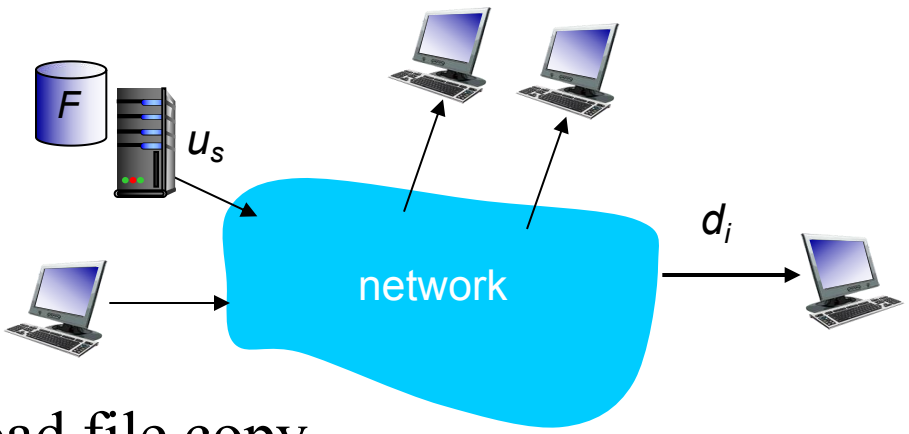
Question: How much time to distribute file (size F) from one server to N *peers*?

- peer upload/download capacity is limited resource
- **Distribution time:** the time it takes to get a copy of the file to all N peers.



File distribution time: client-server

- Server transmission: must sequentially send (upload) N file copies:
  - time to send one copy: F/u_s
  - time to send N copies: NF/u_s



- Client: each client must download file copy
  - d_{min} = min client download rate
  - maximum client download time: F/d_{min}

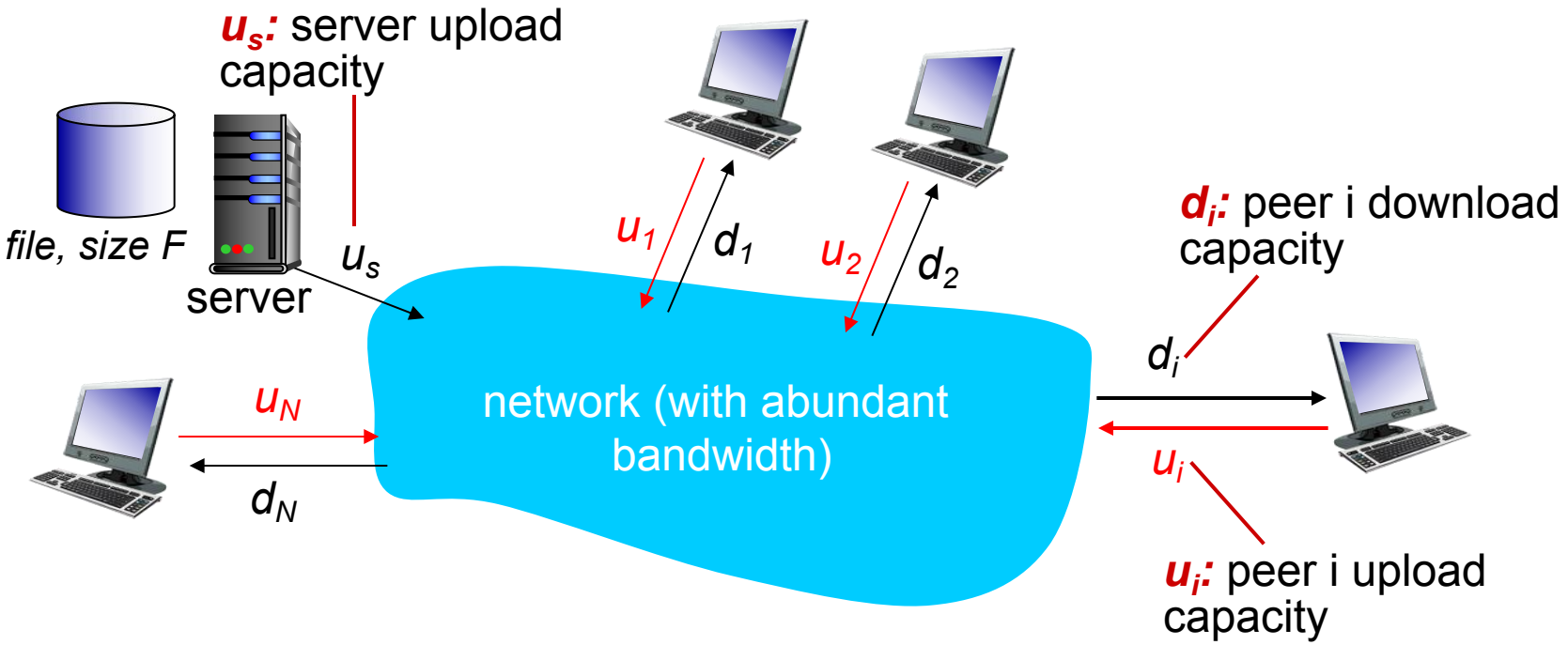
time to distribute F to N clients using client-server approach

$$D_{c\text{-}s} \geq max\{NF/u_s, F/d_{min}\}$$

increases linearly in N

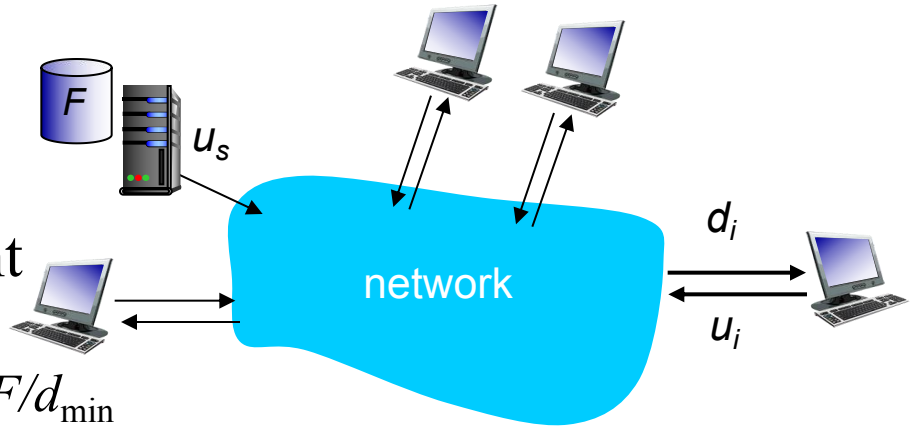
File distribution time: P2P

In P2P model, clients are both downloaders and uploaders.



File distribution time: P2P

- Server transmission: must upload at least one copy
  - time to send one copy: F/u_s
- Client downloading: each client must download file copy
  - maximum client download time: $F/d_{\min}$
- Clients and server: delivering a total of NF bits
  - max upload rate (limiting max download rate) is $u_s + \Sigma u_i$



time to distribute F to N clients using P2P approach

$$D_{P2P} \geq \max\{F/u_s, F/d_{min}, NF/(u_s + \Sigma u_i)\}$$

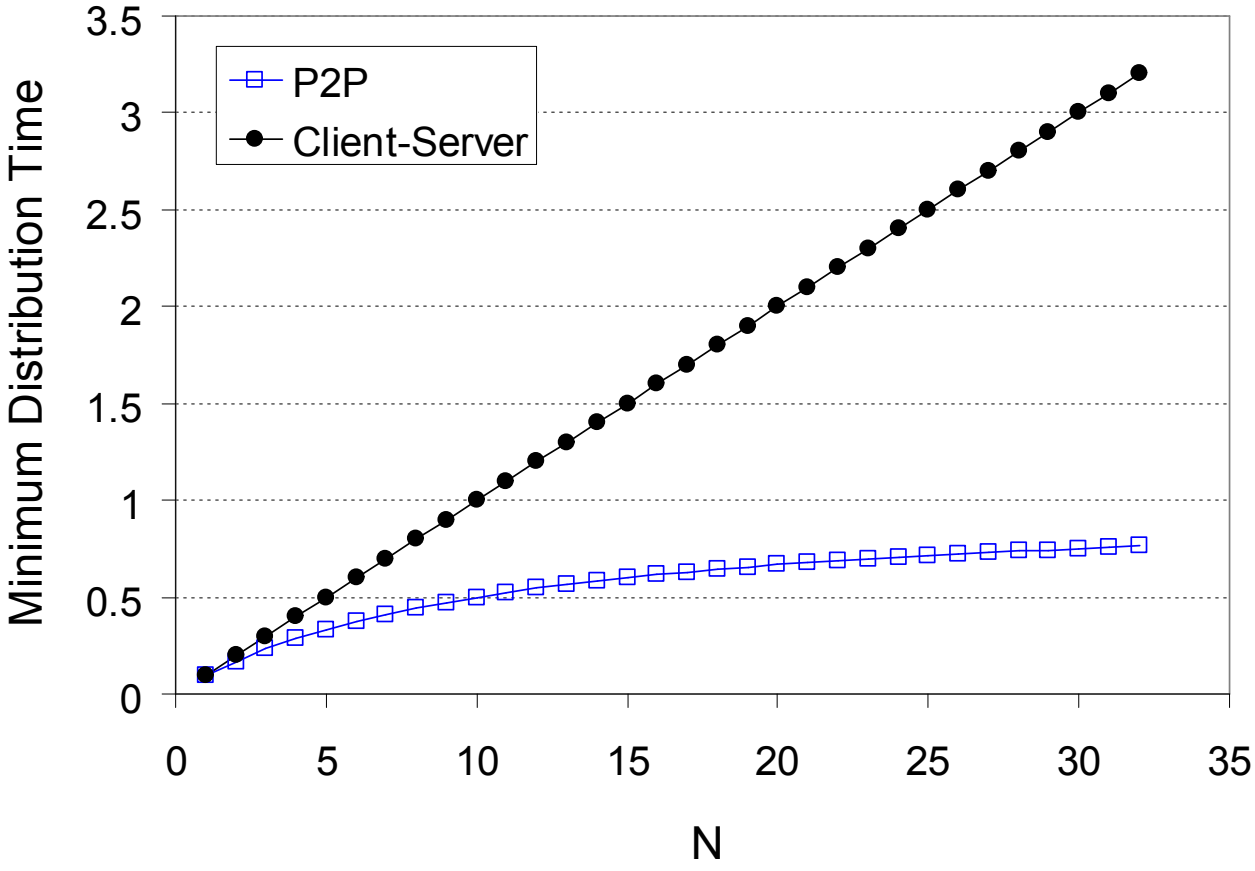
If each peer can redistribute a bit as soon as it receives the bit, then there is a scheme that actually achieves this lower bound

increases linearly in N …

… but so does this, as each peer brings service capacity

Client-server vs. P2P: example

client upload rate = u, F/u = 1 hour, $u_s = 10u$, $d_{min} \geq u_s$



DNS Overview

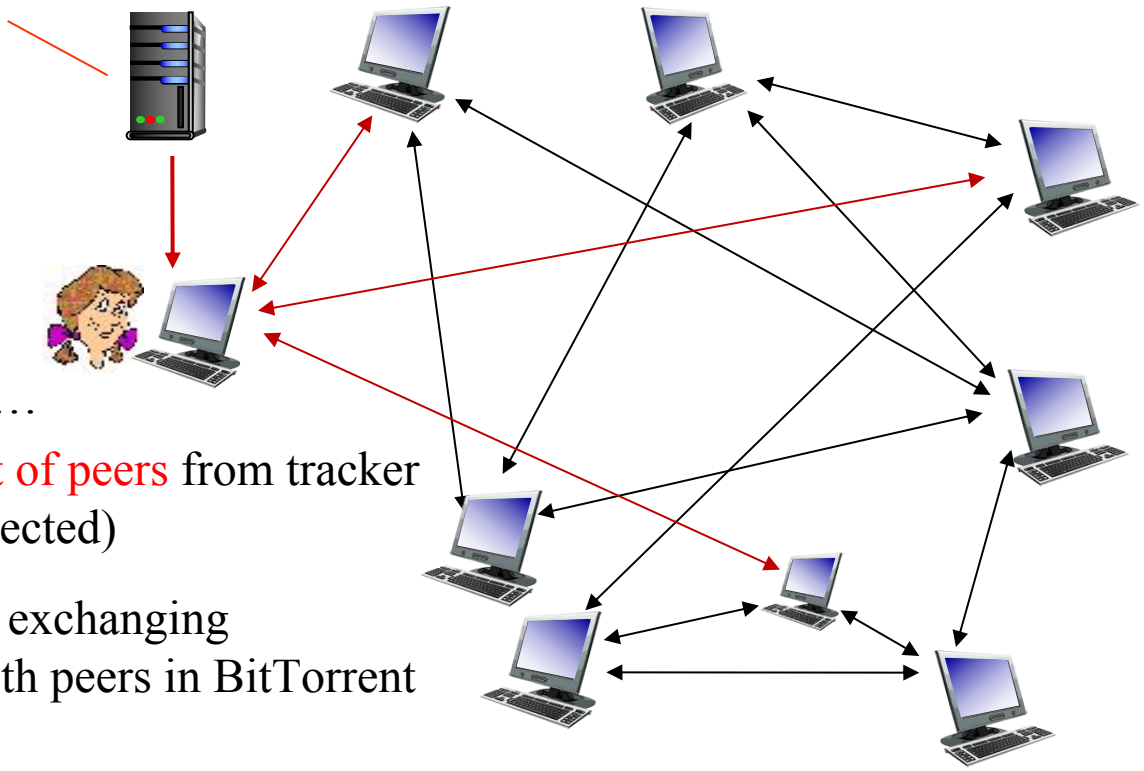
- P2P vs Client Server
- BitTorrent

P2P file distribution: BitTorrent

- File divided into 256Kb chunks
- Peers in BitTorrent send/receive file chunks

tracker: tracks peers participating in BitTorrent

torrent: group of peers exchanging chunks of a file



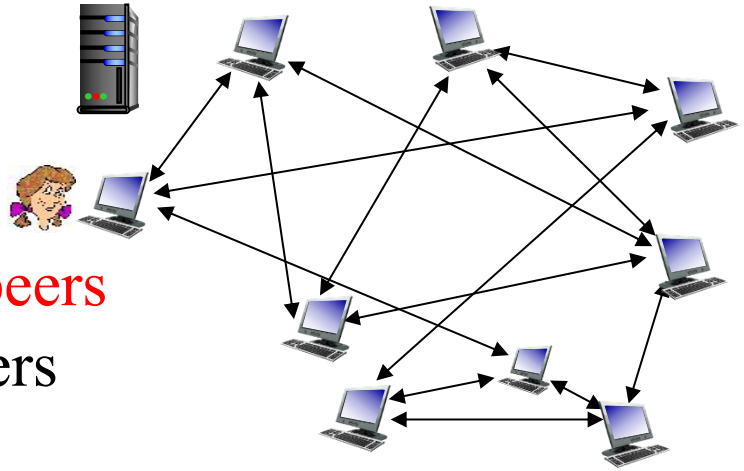
Alice arrives …

… obtains list of peers from tracker (randomly selected)

… and begins exchanging file chunks with peers in BitTorrent

P2P file distribution: BitTorrent

- Peer joining BitTorrent:
  - has no chunks, but will accumulate them over time from other peers
  - registers with tracker to get list of peers
  - TCP connections with subset of peers ("neighbors")



- While downloading, peer uploads chunks to other peers
  - Peers may leave
  - Peers may come, initiating connections with Alice

- Once peer has entire file, it may (selfishly) leave or (altruistically) remain in BitTorrent

BitTorrent: requesting, sending file chunks

Q1: which chunks should she request first from her neighbors?

Q2: to which of her neighbors should she send requested chunks?

requesting chunks:

- at any given time, different peers have different subsets of file chunks
- periodically, Alice asks each "neighbor" for list of chunks that they have
- Alice requests missing chunks from peers, rarest first

sending chunks: tit-for-tat

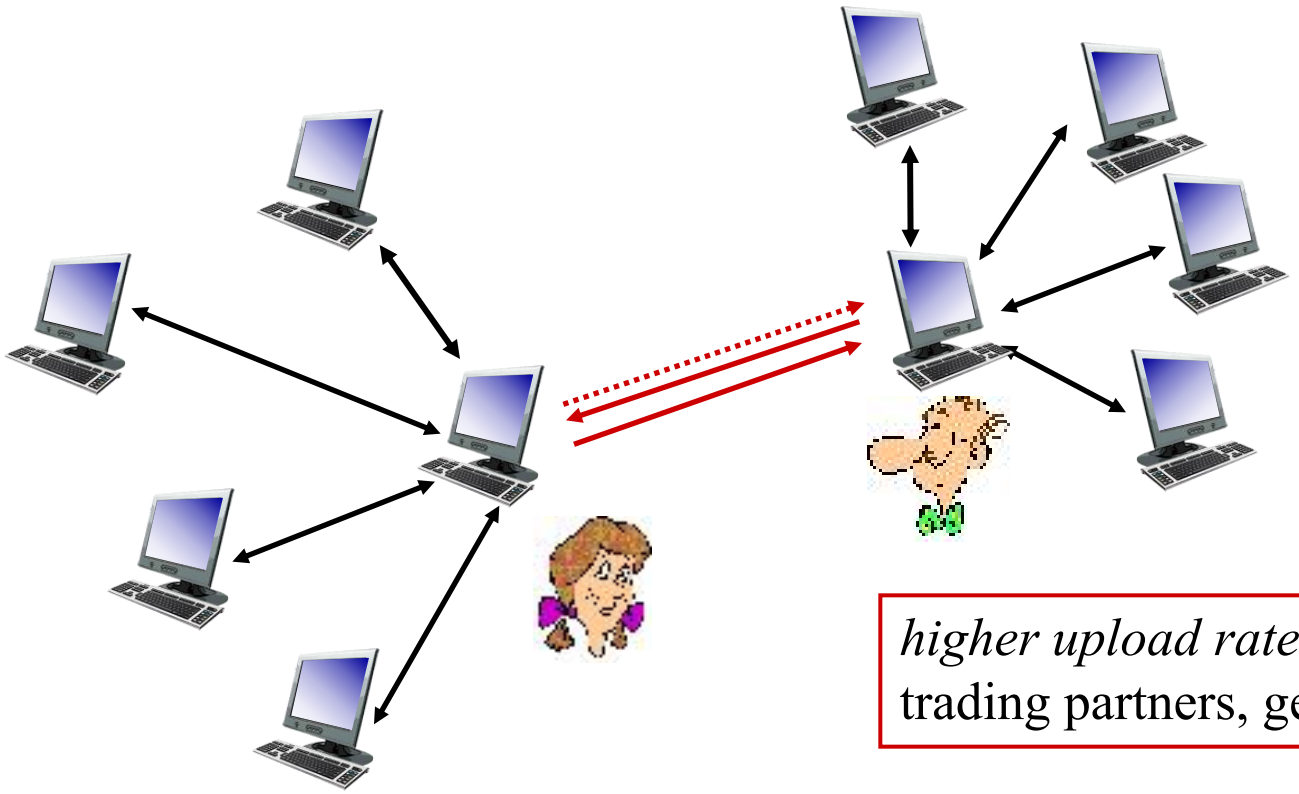
- Alice sends chunks to those four peers currently sending her chunks at highest rate
  - other peers are choked by Alice (do not receive chunks from her)
  - re-evaluate every 10 secs
- every 30 secs: randomly select one additional peer, starts sending chunks
  - "optimistically unchoke" this peer
  - newly chosen peer may join top 4

BitTorrent: tit-for-tat

(1) Alice "optimistically unchokes" Bob

(2) Alice becomes one of Bob's top-four providers; Bob reciprocates

(3) Bob becomes one of Alice's top-four providers



higher upload rate: find better trading partners, get file faster!

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